Metric algebraic critical point theory

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1 Generalities on Lipschitz Functions

1.1 Lipschitz functions on manifolds and subdifferentials

1.1.1 General facts and definitions

For all the following, M will denote a differentiable manifold of dimension m. In this section, we generalize many definitions related to Lipschitz functions. We begin by extending the definition of Lipschitz functions to functions on a manifold.

Definition 1. *A function $f : M \to \mathbb{R}^n$, where M is a smooth manifold, is locally Lipschitz if for any chart $U \subset M$ and any compact $K \subset U$, there exists $C > 0$ such that, with the coordinates on U*

$$\forall x, y \in K, |f(x) - f(y)| \leq C|x - y|$$

Let $f : U \subset \mathbb{R}^m \to \mathbb{R}^n$ be a Lipschitz function, where U is open. Rademacher's theorem, which states that f is differentiable almost everywhere, will allow us to generalize differentials to this setting.

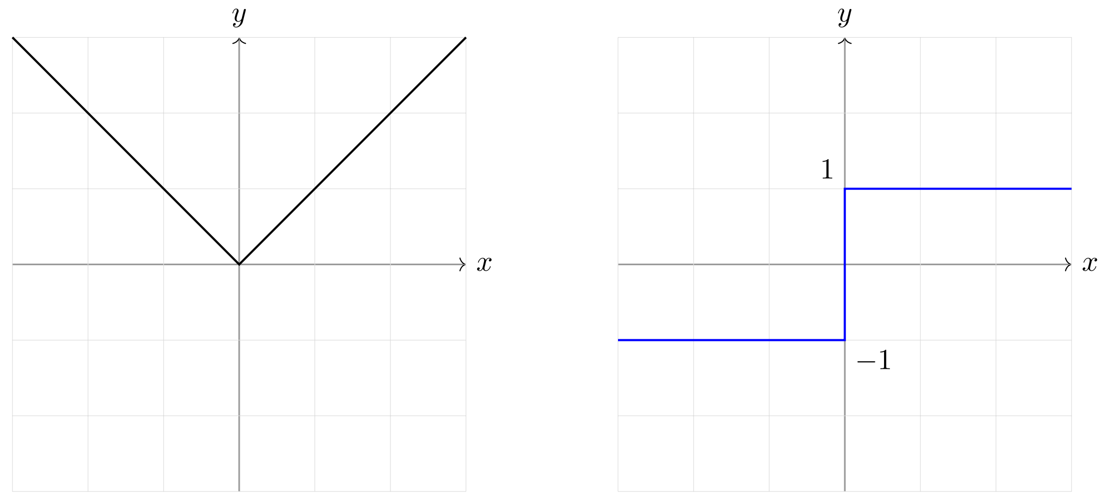
Definition 2. *[10] The subdifferential of f at x, denoted by $\partial_x f$ is the convex hull of the linear maps l obtained as*

$$l = \lim_{i\to\infty} d_{x_i} f$$

where f is differentiable at each x_i and $x_i \to x$.

The subdifferential is then set valued. This set is always non empty and compact, f being Lipschitz bounds df where it is defined.
By [9], we can exclude a set of measure 0 from the possible values of x_i without changing the definition.

Figure 1: The absolute value and its subdifferential represented in the plane
The set on the right is $\bigcup_{l\in\partial f} l$



Proposition 1. *∂f is the smallest convex valued upper semi-continuous function containing df whenever it is defined.*

Proof. First, ∂f is upper semi-continuous : by compactness, it suffices to show that for all $x \in U, \varepsilon > 0$, for y sufficiently close to x.

$$\partial_y f \subset \partial_x f + B(0, \varepsilon)$$

Assume for contradiction that there exists a sequence $x_i \to x$ such that for all i, $\partial_{x_i} f$ has an element out of $A := \partial_x f + \overline{B}(0, \varepsilon/2)$. A being convex and closed, let x'_i be such that $d_{x'_i} f$ is out of A, and such that $x'_i - x_i \to 0$. Up to extraction, $d_{x'_i} f$ converges to an element in $\partial_x f$, so it enters A, contradiction.

The minimality of ∂f follows from the fact that it is a convex hull. $\square$

The subdifferential allows us to generalize the idea of regular and critical points, which will be an important tool for generalizing Morse theory.

Definition 3. *A point $x \in U$ will be called critical if there exists some $l \in \partial_x f$ not of maximal rank. x will be called regular if it isn't critical.*

To extend these subdifferentiability notions to manifolds, one must ensure that for all charts, the definitions of ∂f coincide. If this is the case, $\partial_x f$ is a compact convex subset of $T_x^* M$. The following generalization of the chain rule allows us to do that.

Proposition 2 (Subdifferential chain rule)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}^k$ be a C^1 function. For $x \in U$:*

$$\partial_x (h \circ f) \subset d_{f(x)} h \circ \partial_x f$$

If h is a diffeomorpism, we have equality.

Proof. For the first part, because f is differentiable a.e, it suffices to show that the leftwise term is an upper semi-continuous convex valued function that contains $d(h \circ f)$ whenever df is defined. This follows from the definition of ∂f.

If h is a diffeomorphism, then for $x \in U$:

$$\partial_x f = \partial_x (h^{-1} \circ h \circ f) \subset d_{h \circ f(x)} h^{-1} \circ \partial_x (h \circ f) \subset \partial_x f$$

so we have equalities, and the result follows. □

Although some important differentiablity related results can be obtained with locally Lipschitz functions as we'll see, a general locally Lipschitz function can have some pathological behaviours. As an example, it's subdifferential isn't in general a topological submanifold of M : define

$$A = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+1}}, \frac{1}{2^{2n}} \right[, B = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+2}}, \frac{1}{2^{2n+1}} \right[$$

The integral of the following function constitutes a counterexample, no chart can be found around the point $(0, 1)$ which is a point in the subdifferential.

$$\begin{aligned} \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto \begin{cases} 2 & x \in A \\ 1 & x \in B \\ 0 & x \leq 0 \text{ or } x \geq 1 \end{cases} \end{aligned}$$

However, the functions we'll study, for example continuous selections (see Section 2) have much nicer behaviour and we retrieve this property.

1.1.2 The case of convex functions

If we consider convex Lipschitz functions $f : U \subset \mathbb{R}^m \to \mathbb{R}$, where U is a convex open set, the subdifferential can be understood much more precisely (in fact, this object was first invented for convex functions).

Consider a point $x \in U$ where $d_x f$ exists. By convexity, we have

$$\forall z \in U, f(z) \geq f(x) + d_x f(z - x)$$

Denote C_x the set of all linear maps verifying such property at a point x. The function $C : x \mapsto C_x$ is convex-valued, upper semi-continuous and contains $d_x f$ whenever it is defined. Moreover, at such points we have for h small enough :

$$f(x + h) = f(x) + d_x f(h) + o(h) \geq f(x) + \varphi(h)$$

where $\varphi \in C_x$. We deduce $(\varphi - d_x f)(h) = o(h)$, so $\varphi = d_x f$. Upper semi-continuity gives us $C \subset \partial f$, and the reciprocal is given by minimality of ∂f. So for a convex function the subdifferential can be described as

$$\partial_x f = \{\varphi \in (\mathbb{R}^m)^* \mid \forall z \in U, f(z) \geq f(x) + \varphi(z - x)\}$$

which gives a geometrical meaning to the name subdifferential.

Lastly, we connect the subdifferential to the directionnal derivatives of f. The directionnal derivative of f at x in direction $v \in \mathbb{R}^m$ is given by

$$D_x f(v) = \lim_{t \to 0^+} \frac{f(x + tv) - f(x)}{t}$$

and is always defined since f is convex. The two concepts are related by the following formula

$$D_x f(v) = \max_{l \in \partial f} \langle l, v \rangle$$

1.2 Generalized differential theorems

We now consider the specific case of $f : U \subset \mathbb{R}^m \to \mathbb{R}^m$. This section is devoted to proving generalizations by Clarke [10] of the inverse function theorem and the implicit function theorem for the case of Lipschitz functions.

In this section, we norm the space of linear maps $\mathbb{R}^m \to \mathbb{R}^m$, which we denote $\mathcal{L}(\mathbb{R}^n)$, with the operator norm $\|\cdot\|$ obtained from the norm used on $\mathbb{R}^n$.

Lemma 1. *Assume U is convex, and that there exists some closed convex set C such that*

- *Every $l \in C$ is invertible.*
- *$\partial_x f \subset C$ for all $x \in U$*

Then there exists some $\delta > 0$ such that for all $x, y \in U$:

$$|f(x) - f(y)| \geq \delta |x - y|$$

Proof. Firstly, we set

$$\delta = \inf \{|l(h)| \mid l \in C, h \in S\}$$

where S is the unit sphere in $\mathbb{R}^n$. $\delta > 0$ because all $l \in C$ are invertible and $C \times S$ is closed.

Let $x, y \in U$ and assume $x \neq y$. Let π be the plane perpendicular to $y - x$ passing through x. f being differentiable a.e, Fubini's theorem implies that for almost all $x' \in \pi$, f is differentiable almost on the whole line passing through x' and perpendicular to π (i.e f isn't differentiable on a subset of this line of one dimensional measure 0).
Denoting $F : t \longmapsto f(x' + t(y - x))$, we have :

$$f(x' + y - x) - f(x') = \int_0^1 F'(t)dt = \int_0^1 d_{x'+t(y-x)} f(y - x)dt \tag{1}$$

Define the linear map

$$L = \int_0^1 d_{x'+t(y-x)} f dt$$

C being convex and closed, $L \in C$. Observe that the integral on the right of (1) is just $L(y - x)$. So

$$|f(x' + y - x) - f(x')| \geq \delta |y - x|$$

This inequality holds for almost all $x' \in \pi$, so by continuity of f, it holds for x and we get the desired result. □

Theorem 1 (Lipschitz inverse function theorem)**.** *If x_0 is a regular point of f, then there exists V, W, neighbourhoods of x_0 and $f(x_0)$ respectively, such that f induces a bijection $f : V \to W$ where f^{-1} is Lipschitz too.*

Proof. By compactness of $\partial_{x_0} f$, regularity of x_0, and the fact that the invertible linear maps form an open set, there exist some $\varepsilon > 0$ such that the set

$$C := \partial_{x_0} f + \overline{B}(0, \varepsilon)$$

is composed only of invertible maps. By upper semi-continuity of ∂f, there is some $r > 0$ such that for $y \in B(x_0, r)$, $\partial f(y) \subset C$. We then apply Lemma 1 with $U = B(x_0, r)$ and get a corresponding $\delta > 0$.

Set $V = B(x_0, r/2)$ and $W = f(V)$. We showed in particular that f is injective on $\overline{V}$, by compactness $f : \overline{V} \to f(\overline{V})$ induces a homeomorphism. W is then open. The inequality we got with δ implies that f^{-1} is $\frac{1}{\delta}$ Lipschitz. □

From this theorem, follows the implicit function theorem for Lipschitz functions

Theorem 2. *Let $F : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^m \times \mathbb{R}^n$ is open and F is Lipschitz. Let $(x_0, y_0) \in U$ such that $F(x_0, y_0) = 0$ and y_0 is a regular point of $y \mapsto F(x_0, y)$. Then there exist neighbourhoods $(x_0, y_0) \in W \subset U$, $x_0 \in V \subset R^m$ and a Lipschitz function $g : V \to \mathbb{R}^n$ such that*

$$(x, y) \in W \text{ and } F(x, y) = 0 \Leftrightarrow x \in V \text{ and } y = g(x)$$

The proof is almost the same as the usual implicit function theorem, where the continuity of the differential is replaced by the semi continuity of ∂F.

1.3 First Morse lemma for locally Lipschitz functions

The goal of this section is to use the generalized notion of critical points in order to generalize the first Morse lemma to the case of locally Lipschitz functions. Fix M an m-dimensional smooth manifold, $f : M \to \mathbb{R}$ a continuous function and define for all $c \in \mathbb{R}$ the sublevel set of f of value c as

$$M_c := f^{-1}(]-\infty, c])$$

The classical first Morse lemma states that if f is C^1, and there are no critical points in $f^{-1}([a, b])$ (that we assume to be compact), then M_b is diffeomorphic to M_a and M_a is a deformation retract of M_b. We extend this to locally Lipschitz functions, following [1]. Note that in this setting x is a critical point of f iff $0 \in \partial_x f \subset T_x^* M$.

Proposition 3. *Let $a < b \in \mathbb{R}$ such that $f^{-1}([a, b])$ is compact and free of critical points. Then there exists a locally Lipschitz homotopy $H : M_b \times [0, 1] \to M_b$ such that for all $s \in [0, 1]$, $H(\cdot, s)$ is a homeomorphism verifying*

$$H(M_b \times \{s\}) \subset M_{b(1-s)+as}$$

Proof. We'll construct $V \in \mathrm{Vec}(M)$ a smooth complete vector field on M such that, for all $x \in M$ such that $d_x f$ is defined :

$$d_x f(V(x)) \leq 0 \text{ and } d_x f(V(x)) \leq -1 \text{ if } x \in f^{-1}([a, b]) \tag{2}$$

Let $x \in f^{-1}([a, b])$. x being regular and $\partial_x f$ convex, we find an affine hyperplane in $T_x^* M$ separating it from 0. This gives us $v_x \in T_x M$ such that $\langle \partial_x f, v_x \rangle \leq -2$. By semi continuity, there exists an open neighbourhood U_x of x such that for all $y \in U_x$, $\langle \partial_y f, v_x \rangle \leq -1$. We choose this neighbourhood such that it has compact closure.

These U_x cover $f^{-1}([a, b])$. Let $U_{x_1}, ..., U_{x_r}$ be a finite subcover, and let $\rho_1, ..., \rho_r$ be a partition of unity subordinated to this covering. We set

$$\forall x \in M, V(x) = \sum_{i=1}^{r} \rho_i(x) v_{x_i}$$

which indeed verifies (2) : if $d_x f$ is defined, it is contained in $\partial_x f$, and V is of compact support so it is complete.

Now observe that for $x \in f^{-1}([a,b])$, $t \mapsto f(e^{tV}x)$ is strictly decreasing. Denote by $t(x,s)$, $s \in [0,1]$, the unique time such that $f(e^{t(x,s)V}x) = as + f(x)(1-s)$. The equation defining $t(x,s)$ is locally Lipschitz in s, x and t, so by the Lipschitz implicit function theorem, t is a locally Lipschitz function of (x,s).

We can similarly define t' a locally Lipschitz function of x such that $f(e^{t'(x)V}x) = b$ for x in the orbit of some $y \in f^{-1}(b)$ by V. Denote O the set of $x \in M$ that are in such orbits. The homotopy is then given by

$$H : \begin{array}{cl} M_b \times [0,1] & \longrightarrow M_b \\ (x,s) & \longmapsto \begin{cases} e^{t(e^{t'(x)V}x,s)V}x & \text{if } x \in O \\ x & \text{else} \end{cases} \end{array}$$

$\square$

2 Morse theory of continuous selections

2.1 Definitions

We now restrict ourselves to continuous functions that are piecewise smooth with finitely many pieces. We define those as continuous selections.

Definition 4. *A continuous function $f : M \to \mathbb{R}$ is a continuous selection if there exists $\varphi_1, ..., \varphi_n$ "regular" functions such that*

$$\forall x \in M, f(x) \in \{\varphi_1(x), ..., \varphi_n(x)\}$$

The term "regular" is used loosely here to indicate that the φ_i satisfy some extra condition on top of being continuous. For what follows, we'll restrict ourselves to continuous selections of smooth functions (but C^2 will suffice).

Note that a continous selection of locally Lipschitz functions is itself locally Lipschitz : if M is an open set of $\mathbb{R}^m$, and we assume that the functions φ_i are Lipschitz with constants c_i, then f is Lipschitz with constant $\max_i c_i$. Because smooth functions are locally Lipschitz, the functions we'll study will be too.

We can then relate ∂f to the differentials of the φ_i. Indeed, a limit of points of the form $d_x\varphi_i$ in T^*M is of the same form. Define the set of active indices at x,

$$I(x) = \{i \in [\![1, n]\!] \mid \varphi_i(x) = f(x)\}$$

$\partial_x f$ is then the convex hull of some $d_x\varphi_i$, $i \in I(x)$. It is possible that not all $d_x\varphi_i$ are used, if for example one of the functions isn't active in a neighbourhood of x.

We would like to define what it means for such a continuous selection to be Morse. Let f be a continuous selection of $\varphi_1, ..., \varphi_n$. Following [1], we give three conditions for a function to be considered Morse.

$$\underline{\text{The graphs of the } \varphi_i, i \in [\![1, n]\!], \text{ are transversal}} \tag{C1}$$

This condition has some important consequences : for a non empty $I \subset [\![1, n]\!]$, we define

$$W_I := \{x \in M \mid I(x) = I\}$$

Condition (C1) implies that W_I is a submanifold of M of codimension $|I| - 1$: the intersection of the graphs of $\varphi_i, i \in I$ is a submanifold of $M \times \mathbb{R}$ of codimension $|I|$ and W_I is the pre-image of this intersection through the diffeomorphism $x \mapsto (x, \varphi_i(x))$, $i \in I$, minus the closed set $\bigcup_{J \supset I} W_J$.

This transversality condition also implies that for $x \in M$, the differentials $d_x\varphi_i$ are affinely independant where $i \in I(x)$. By affinely independant, we mean that one of the two equivalent conditions are satisfied : for some/any $j \in I(x)$, the family $(d_x\varphi_i - d_x\varphi_j)_{\substack{i \in I(x) \\ j \neq i}}$ is linearly independant, or equivalently, the family

$$\begin{pmatrix} d_x\varphi_i \\ 1 \end{pmatrix}, i \in I(x)$$

is linearly independant. This implies that $\partial_x f$ is a simplex with less than $|I(x)| - 1$ vertices.

Note that for all $I \subset [\![1, n]\!]$, $f_{|W_I}$ is a smooth function. This motivates the second condition

$$\underline{\text{For all non empty } I \subset [\![1, n]\!], f_{|W_I} \text{ is Morse}} \tag{C2}$$

Consider $x \in M$ a critical point of f. Observe that x must also be a critical point of $f_{|W_{I(x)}}$. So

$$T_x W_{I(x)} \subset \bigcap_{i \in I(x)} d_x\varphi_i$$

but the converse can fail if the $d_x\varphi_i$ are not linearly independant enough. If this intersection is indeed larger, the geometry of the graph of f does not correspond to what we intuitively picture as a "non degenerate critical point" (see figure 2 for a 1 dimensional example). We then impose the following condition, ensuring equality.

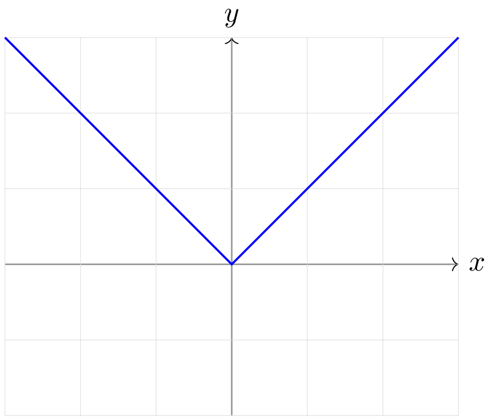
$$\underline{\forall x \in M, \forall j \in I(x), (d_x\varphi_i)_{\substack{i\in I(x)\\ i\neq j}} \text{ are linearly independant}} \tag{C3}$$

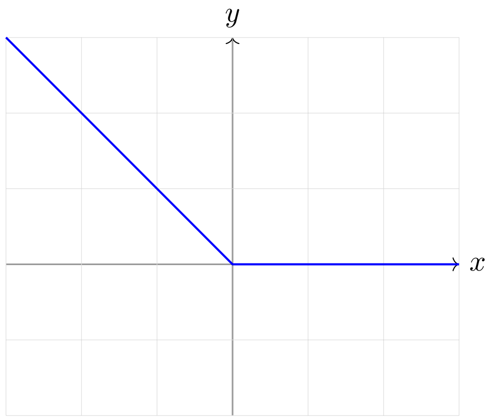
Geometrically, this condition corresponds to requiring that 0 is either outside or in the interior of the simplex $\operatorname{conv}\{d_x\varphi_i \mid i \in I(x)\}$, but not on one of its faces. Note that this condition can be relaxed by restricting it to the subset of $I(x)$ participating in $\partial_x f$.

A continuous selection f is then called a Morse function if it verifies all three conditions. We can also impose that these conditions only hold at critical points, and we get the same theory.

Note that as in the smooth case, a generic continuous selection is Morse, i.e the set of all n-tuples $(\varphi_1, ..., \varphi_n)$ such that these conditions hold at every critical point is open dense in C^2 topology [13]. We can see it intuitively by noticing that (C1) is a generic condition, any two functions can be slightly perturbed in order for their graphs to be transversal, and any small perturbation of transversal graphs preserves transversality. By genericity of smooth Morse functions, (C2) is also a generic condition. With the geometric interpretation, we can also see that condition (C3) is generic.

Figure 2: Example and counterexample of condition (C3)





2.2 Normal form of a Morse continuous selection

We show here an equivalent of the second Morse lemma for the case of Morse continuous selections. Noticing that $x \mapsto I(x)$ is upper semi-continuous and using the definitions yields the following result.

Lemma 2. *If $f : M \to \mathbb{R}$ is a Morse continuous selection, all its critical points are isolated.*

We'll then consider, up to restricting ourselves to a small enough chart, that $f : \mathbb{R}^m \to \mathbb{R}$ is Morse and has no critical points other than 0. We'll write for shorthand $W := W_{I(0)}$, $k := |I(0)| - 1$, $c := f(0)$, and assume that $I(0) = [\![1, k+1]\!]$. Semi-continuity of the active index also allows us to assume that for $x \in \mathbb{R}^m$, $I(x) \subset [\![1, k+1]\!]$.

Notice that because of condition (C1) and semi-continuity of the active index, the family $(W_I)_{I\subset[\![1,k+1]\!]}$ gives a Whitney stratification of $\mathbb{R}^m$. This observation motivates the following lemma, in the spirit of Thom isotopy theory.

Lemma 3. *Let $h : \mathbb{R}^k \times \mathbb{R} \to \mathbb{R}$ be a continuous selection of at most $k+1$ smooth functions verifying condition* (C1)*. There exists a neighbourhood U of* 0 *and $\varepsilon > 0$, a vector field $X : U \times]-\varepsilon, \varepsilon[\to \mathbb{R}^k \times \mathbb{R}$ verifying*

1. For $x \in U, t \in]-\varepsilon, \varepsilon[$ we have

$$X(x,t) = (\tilde{X}(x,t), 1)$$

2. If $(x,t) \in W_I$ for some active index set I, then X is tangent to W_I at all times and the restriction to W_I is C^1

3. $(x,t) \mapsto h(x,t)$ is constant along integral curves of X

4. The flow of X is continuous.

The use of this lemma requires a parameter dependent continuous selection. For the proof of the main theorem, we will have to build one with some boundary conditions wich is done in the following way

Lemma 4. *Let $\varphi_1, ..., \varphi_{k+1} : \mathbb{R}^k \times \mathbb{R} \to \mathbb{R}$ be smooth functions verifying conditions* (C1) *and of value* 0 *on $\{0\}^k \times \mathbb{R}$. Let $h^0 : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $\varphi_1(\cdot, 0), ..., \varphi_{k+1}(\cdot, 0)$. Then there exist a neighbourhood U of 0 and a continuous selection $h : U \times [0,1] \to \mathbb{R}$ of $\varphi_1, ..., \varphi_{k+1}$ such that $h(\cdot, 0) = h^0$.*

Proof. For all $i < k+1$ define $g_i(x,t) = \varphi_i(x,t) - \varphi_{k+1}(x,t)$. By affine independance the family $(d_0 g_i)_i$ is linearly independant. Using the implicit function theorem, there are coordinates on a neighbourhood $U \times J$ of 0 linearizing all g_i, while sending $\mathbb{R}^k \times \{0\}$ to a hyperplane. We assume without loss of generality that this hyperplane is $\mathbb{R}^k \times \{0\}$. It follow that for every i, j, $\varphi_i - \varphi_j$ is a linear function in these new coordinates.

Denote $W_I \subset \mathbb{R}^k \times \{0\}$ the set of points $(x, 0)$ such that the active index of h^0 at x is equal to $I \subset [\![1, k+1]\!]$. We also define $L = \{g_i = 0 \mid i \in [\![1, k+1]\!]\}$ which is a linear subspace of dimension 1. If we now define for all I, $C_I = W_I + L$, it defines a partition of $\mathbb{R}^{k+1}$.

Observe that for $z = (x, 0) + u$, $(x, 0) \in W_I, u \in L$ we have for $i, j \in I$ that

$$\varphi_i(z) - \varphi_j(z) = \varphi_i(x, 0) - \varphi_j(x, 0) = 0$$

by linearity. We can then define for all $(x,t) \in C_I$ the function $h(x,t) = \varphi_i(x,t)$ for any $i \in I$. This h defines a continuous selection coinciding with h^0 on $\mathbb{R}^k \times \{0\}$ in the new coordinates.

We now want to extend this selection on the interval $[0, 1]$. By repeating the argument for some starting time different than 0, connectedness and compactness yield that there is a neighbourhood U of 0 such that we can define h on $U \times [0, 1]$. $\square$

We are now ready to prove the main result of this section.

Theorem 3 (Normal form)**.** *If 0 is a critical point of f, then there exists continuous local coordinates around 0, $\psi : \mathbb{R}^k \times \mathbb{R}^{m-k} \to U$ such that*

$$\forall y \in \mathbb{R}^k, \forall z \in \mathbb{R}^{m-k}, f(\psi(y,z)) = \lambda(y) + q(z)$$

where λ is a continuous selection of $k+1$ affinely independant linear functions, and q is of the form

$$\forall z \in \mathbb{R}^{m-k}, q(z) = c - \sum_{i=1}^{\mu} z_i^2 + \sum_{i=\mu+1}^{m-k} z_i^2$$

with $0 \leq \mu \leq m-k$.

Proof. We first restrict ourselves to O a neighbourhood of 0 such that W can be considered rectified in some smooth coordinates i.e, $W \cap O = 0 \times \mathbb{R}^{m-k}$. By (C2) we can also use the smooth Morse lemma to get $f(0,z) = q(z)$. For $(y,z) \in \mathbb{R}^m$, write $f^0(y,z) = f(y,z) - f(0,z)$. We will build a locally invertible map $\psi : \mathbb{R}^m \to \mathbb{R}^m$ such that $\psi(0) = 0$ and for (y,z) close enough to 0, $f^0 \circ \psi(y,z)$ is a continuous selection of linear maps independant of z.

Assume that all φ_i are of value 0 at 0, up to replacing f by f^0. We will show that up to coordinate change, f^0 is only a function of y and not of z. This can be done in the following way : first, by condition (C3), we can assume that the k first φ_i are linearized, i.e equal to dy_i in a neighbourhood of 0. The last φ_{k+1} could still depend on z.

To fix this we apply lemma 3 to $f^0 : \mathbb{R}^{m+k-1} \times \mathbb{R} \to \mathbb{R}$. Denote $\Phi(s, y, z)$ the point obtained by integrating X for time s starting at (y, z). The following local homeomorphism

$$\phi(y, z) = \Phi(-z_{m-k}, y, z) + (0, ..., 0, z_{m-k}) \tag{3}$$

verifies $f^0 \circ \phi^{-1}(y, z) = f^0(y, z_1, ..., z_{m-k-1}, 0)$. By repeating the process we eliminiate the dependance in z.

Note that in the new coordinates, $\varphi_i = dy_i$ for $i < k+1$. To write f^0 as a continuous selection of linear functions, it suffices to linearize φ_{k+1}. For that, for all $t \in [0, 1]$ define

$$\varphi_{k+1}^t = (1-t)\varphi_{k+1} + td_0\varphi_{k+1}$$

Note that for y close enough to 0 and any $t \in [0, 1]$, $d_y\varphi_{k+1}^t$ is close to $d_y\varphi_{k+1}$, hence the functions $dy_1, ..., dy_k, \varphi_{k+1}^t$ verify (C1). By lemma 4, we define $h(y, t)$ for all y close enough to 0 and $t \in [0, 1]$ as a continuous selection of $dy_1, ..., dy_k, \varphi_{k+1}^t$ such that it equals f^0 at $t = 0$.

We then apply lemma 3 and get U an even smaller neighbourhood and X a vector field on a neighbourhood of $(0, 0)$. Up to shrinking U, we can assume that for all t^* there is a time $s(t^*) > 0$ such that for all $y \in U$, we can integrate (y, t^*) for time s, $|s| < s(t^*)$. This gives a covering of $[0, 1]$ by intervals of the form $]t^* - s(t^*), t^* + s(t^*)[$, so we can still cover it with finitely many of these intervals.

Assume for simplicity that we can flow for time 1 from all $(y, 0)$, $y \in U$ i.e $[0, 1]$ can be covered with only one such interval. We construct a local homeomorphism ϕ similarly to (3) such that

$$f^0(y) = h(y, 0) = h \circ \phi^{-1}(y, 1)$$

so f^0 can be written in some coordinates as a continuous selection of linear functions. If we need more than one interval to cover $[0, 1]$ we can decompose this into finitely many homeomorphisms going from 0 to 1 in successive steps.

By composing all the steps, we indeed constructed the local homeomorphism ψ.

□

We now try to understand better the structure of the non smooth component that appears in the normal form theorem. It turns out that it can be encoded in a purely combinatorial way, as the following theorem shows [16].

Theorem 4. *Let $\lambda : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $k+1$ affinely independant linear functions, $\lambda_1, ..., \lambda_{k+1}$. Then there exists a unique family $S_1, ..., S_l$ of subsets of $[\![1, k+1]\!]$ verifying that*

- *$S_i \not\subset S_j$, $i \neq j$*
- *λ can be written as*
$$\lambda(y) = \min_i \max_{j \in S_i} \lambda_j(y)$$

Proof. Let's first tackle uniqueness. Let $S_1, ..., S_l$, $D_1, ..., D_r$ be two such distinct families. Assume without loss of generality that $S_1 \neq D_j$. We can also assume that for all $1 \leq j \leq r$, $S_1 \not\supset D_j$. If it is the case, then for all $1 < i \leq l$ we have $D_j \not\supset S_i$ by assumption on S. We can then swap S and D to get what we want.
By affine independance, the following linear system has a solution

$$\lambda_i(y) + \mu = c_i, i \in [\![1, k+1]\!]$$

where the unknowns are y, μ and $c_i = 0$ if $i \in S_1$ and 1 otherwise. It follows that, if we use the representation with D_i, we get $\lambda(y) = 1 - \mu$ and if we use the one with the S_i, $\lambda(y) = -\mu$.

Now for existence. For all permutation $\nu \in \mathfrak{S}_{k+1}$ define the cone

$$K_\nu = \{y \in \mathbb{R}^k \mid \lambda_{\nu(1)}(y) \leq ... \leq \lambda_{\nu(k+1)}(y)\}$$

and the function $s : \nu \in \mathfrak{S}_{k+1} \mapsto i \in [\![1, k+1]\!]$ where i is such that $\lambda_{\nu(i)} = \lambda$ on K_ν. If we assume that for all y and ν we have

$$\lambda(y) \leq \max\{\lambda_{\nu(1)}(y), ..., \lambda_{\nu(s(\nu))}(y)\}$$

then it suffices to consider the family of sets $S_\nu = \{\nu(1), ..., \nu(s(\nu))\}$ and simplify these by removing those who are contained in one another.

We now give a broad idea of the proof of the claim. The result is true by definition on K_ν. For any $y \in \mathbb{R}^k$, set $y' \in K_\nu$ and consider the evolution of the order of λ_i along the line between y and y' : the value of each λ_i evolves at a constant rate. Whenever the current λ_i defining λ encounters another λ_j and λ becomes defined as λ_j afterwards, either λ_j is smaller than λ_i after the encounter, or the opposite. If we denote I the set of indices such that the functions are smaller than λ_j before the encounter, then in either case we see that λ stays smaller than the maximum of $\lambda_{i'}$ $i' \in I$. This concludes the proof because the line starts at K_ν. $\square$

2.3 Homology of sublevel sets and inequalities

Our goal for this section is to describe more precisely the transition that happens as the critical point of f is passed, where f is still a Morse continuous selection defined on $\mathbb{R}^m$ with 0 as its only critical point.

The normal form of f and theorem 4 allows us to fully describe the behaviour of f with purely combinatorial information, a smooth Morse index and a more complicated set representation. We can encode the combinatorial structure that appears in its non smooth part in a way that will give us topological and homotopical informations about the sublevel sets : define the simplical complex $K(S_1^c, ..., S_l^c)$ on $k+1$ vertices whose faces are given by the complements in $[\![1, k+1]\!]$ of the sets S_i describing the non smooth part of f.

Another thing to note is that when put in normal form, f is a quasi homogeneous function, i.e there exists some real numbers $\alpha_1, ..., \alpha_m > 0$ such that for all $t > 0, x \in \mathbb{R}^m$, we have

$$f(t^{\alpha_1}x_1, ..., t^{\alpha_m}x_m) = tf(x)$$

In this case we have $f(ty, t^{1/2}z) = tf(y, z)$ if we assume that $f(0) = 0$, assumption we'll make without loss of generality. For a quasi homogenous function h, we'll denote δ_t the power-scaling map such that $h(\delta_t x) = th(x)$. The following lemmas on quasi homogeneous functions will prove useful for describing the homotopy type of sublevel sets.

Lemma 5. *Let $h : \mathbb{R}^m \to \mathbb{R}$ be a quasi homogeneous locally Lipschitz function having only 0 as a critical point. We have the following homotopy equivalence*

$$(M_1, M_{-1}) \simeq (C_{h^{-1}(-1)}, h^{-1}(-1))$$

where M_1, M_{-1} are the sublevel sets and $C_{h^{-1}(-1)}$ is the cone over $h^{-1}(-1)$.

Proof. Let $\mathbb{S}$ denote the unit sphere. Define a smooth V over $\mathbb{S}$ such that for all $x \in \mathbb{S}$, $V(x) \in \mathbb{R}^m$ and $\langle \partial_x h, V(x)\rangle \leq -1$. Extend V into a vector field on $\mathbb{R}$ such that it commutes with δ_t for all $t > 0$, i.e for $x \in \mathbb{S}$, $V(\delta_t x) = d_x\delta_t(V(x))$. It follows that $s \mapsto h(e^{sV}x)$ is strictly decreasing for all $x \neq 0$.

From this observation, we define similarly to proposition 3, for all $x \in h^{-1}(]-\infty, 0[)$, $p(x)$, the only point of the orbit of x in $h^{-1}(-1)$. We also define for all $x \in h^{-1}(-1), s \in [0, 1[$, $t(x, s)$ such that it verifies

$h(e^{-t(x,s)V}x) = h(x) + s$. If we then denote by $\pi : [0,1] \times h^{-1}(-1) \to C_{h^{-1}(-1)}$ the projection map, where the top of the cone is the point at height $s = 1$, the homotopy equivalence maps are the following

$$F : \begin{array}{ccl} M_1 & \longrightarrow & C_{h^{-1}(-1)} \\ x & \longmapsto & \begin{cases} \pi(0, p(x)) & x \in M_{-1} \\ \pi(h(x)+1, p(x)) & x \in h^{-1}([-1,0]) \\ \pi(1,0) & x \in h^{-1}([0,1]) \end{cases} \end{array}$$

$$G : \begin{array}{ccl} C_{h^{-1}(-1)} & \longrightarrow & M_1 \\ \pi(s,x) & \longmapsto & e^{-t(x,s)}x \end{array}$$

We can see that $F \circ G = \mathrm{id}_{C_{h^{-1}(-1)}}$. $G \circ F$ is homotopic to the identity : we can use V to contract M_{-1} onto $h^{-1}(-1)$, and also to contract $h^{-1}([0,1])$ onto $h^{-1}(0)$. $\square$

If X and Y are two topological spaces, we define the join of X and Y, denoted by $X * Y$, as the quotient space of $X \times Y \times [0,1]$ by the following equivalence relation

$$\forall x, x' \in X, \forall y, y' \in Y, (x, y, 0) \sim (x, y', 0) \text{ and } (x, y, 1) \sim (x', y, 1)$$

Intuitively, we join together the spaces X and Y with all possible segments between the two. Some examples of join include $X * \{y\}$ wich is the cone of X, $X * \mathbb{S}^0$ which is the suspension of X. More generally the join of X with $\mathbb{S}^n$ is an $n+1$-fold suspension.

By using the same ideas as in lemma 5, one can prove the following useful lemma.

Lemma 6. *Let $h : \mathbb{R}^k \times \mathbb{R}^{m-k} \to \mathbb{R}$ be a quasi homogeneous function such that for all $(y,z) \in \mathbb{R}^m$, we have $h(y,z) = a(y) + b(z)$ where a and b are quasi-homogeneous locally Lipschitz with respective exponents $\alpha, \beta > 0$. We have the following homotopy equivalence*

$$h^{-1}(-1) \simeq a^{-1}(-1) * b^{-1}(-1)$$

If we write $f(y,z) = \lambda(y) + q(z)$ as described by the normal form theorem, what remains to be done to understand the relative homology of (M_1, M_{-1}) is to describe $q^{-1}(-1)$ and $\lambda^{-1}(-1)$. The first one has the homotopy type of $\mathbb{S}^\mu$ where μ is the smooth Morse index of f, by classical Morse theory.

For the second, note that we have the following homotopy equivalence $\lambda^{-1}(-1) \simeq \lambda^{-1}(]-\infty, 0[)$. Using the min max form, we get

$$\lambda^{-1}(]-\infty, 0[) = \bigcup_i \bigcap_{j \in S_i} \lambda_j(]-\infty, 0[)$$

Note that each intersection is a convex cone we'll denote by C_i. We'll relate this to the simplical complex $K(S_1^c, ..., S_l^c)$ by using the nerve of the family $C = (C_i)_i$, i.e we'll look at the following set

$$N(C) := \{J \subset [\![1, l]\!] \mid \bigcap_{j \in J} C_j \neq \varnothing\}$$

From the next lemma, it follows that the nerve of $N(C)$ is exactly the one of the family of sets $S_1^c, ..., S_l^c$.

Lemma 7. *Let $J \subset [\![1, k+1]\!]$. J is the whole set if and only if*

$$\bigcap_{j \in J} \lambda_j^{-1}(]-\infty, 0[) = \varnothing$$

Proof. We first show that if $J = [\![1, k+1]\!]$, we indeed get an empty intersection. Recall that 0 is a critical point of f, so it is a critical point of λ. We then have

$$0 \in \partial_0 \lambda \subset \mathrm{conv}\{\lambda_1, ..., \lambda_{k+1}\}$$

So there exists $\mu_1, ..., \mu_{k+1}$ with sum 1 such that $0 = \sum_i \mu_i \lambda_i$. We then see that all λ_i cannot be strictly negative at the same time.

We now show that if $J \neq [\![1, k+1]\!]$, the intersection is not empty. Condition (C3) gives us linear independance of the family $(\lambda_j)_{j \in J}$. The linear system $\lambda_j(y) = -1, j \in J$ then has a solution, where the unknown is y. $\square$

Because every C_i is convex, their intersections are contractible. Applying Leray's nerve theorem, $\lambda^{-1}(-1)$ has the homotopy type of the simplical complex given by the nerve. By virtue of what we discussed, this complex is exactly $K(S_1^c, ..., S_l^c)$. The homotopy type of $f^{-1}(-1)$ is then given by

$$f^{-1}(-1) \simeq K(S_1^c, ..., S_l^c) * \mathbb{S}^\mu$$

By recalling that a join with a sphere is a suspension, we get the following result in homology.

Theorem 5. *For $i \geq 0$, we have*

$$H_{i+\mu}(M_1, M_{-1}) = \tilde{H}_{i-1}(K(S_1^c, ..., S_k^c))$$

Proof. By the previous discussion and by contractibility of the cone, we have for all $i \geq 0$ the exact sequence

$$0 \longrightarrow H_{i+\mu}(C_{h^{-1}(-1)}, h^{-1}(-1)) \longrightarrow \tilde{H}_{i+\mu-1}(h^{-1}(-1)) \longrightarrow 0$$

which yields

$$H_{i+\mu}(C_{h^{-1}(-1)}, h^{-1}(-1)) = \tilde{H}_{i-1+\mu}(K(S_1^c, ..., S_k^c) * \mathbb{S}^\mu)$$

hence by the fact that the join with a sphere is a suspension, we get the desired result. $\square$

We now return to a general manifold M and a continuous selection $f : M \to \mathbb{R}$. We will also assume that f has compact sublevel sets. By excision, the results we got in homology in a chart around a critical point still hold. Then, using the same proofs as with classical Morse theory (cite Milnor), a relation of homology ranks.

For what follows, we will denote by $C(f)$ the set of critical points of f, that is finite. Then for all $x \in C(f)$ and $i \geq 0$, denote by $c_i(x)$ the rank of the homology group $H_i(M_{f(x)+\varepsilon}, M_{f(x)-\varepsilon})$ for some small $\varepsilon > 0$.

Proposition 4. *We have the following relations for all $n \geq 0$*

$$\sum_{i=0}^{n} (-1)^{2n-i} b_{n-i} \leq \sum_{i=0}^{n} (-1)^{2n-i} \sum_{x \in C(f)} c_{n-i}(x)$$

from which we get

$$b_i \leq \sum_{x \in C(f)} c_i(x)$$

3 First properties of the euclidean distance

In this section, we will work in the euclidean space $\mathbb{R}^m$. Some of what we'll do here can be generalized to a more general Riemannian manifold. All throughout, $X \subset \mathbb{R}^m$ will be a smooth manifold unless specified otherwise, $Y \subset \mathbb{R}^m$ a closed set.

The main function we will study is the euclidean distance from Y restricted to X, i.e

$$\forall x \in X, \mathrm{dist}_Y(x) = \min_{y \in Y} |x - y|$$

where $|\cdot|$ is the euclidean norm. This function is locally Lipschitz. In what will follow, we identify $(\mathbb{R}^m)^*$ with $\mathbb{R}^m$ via the euclidean structure. $\partial_x(\mathrm{dist}_{Y|X})$ will sometimes be identified with a subset of T_xX, where $x \in X$.

Our main objectives will be to understand the subdifferential of this function and to find a setting where Morse theory of continuous selections can be applied.

3.1 The subdifferential of the distance function

Our goal in this section will be to compute explicitly the subdifferential of $\mathrm{dist}_{Y|X}$, which gives rise to a very nice and intuitive geometric picture. We first require some results on "infinite continuous selections", which is indeed how dist_Y is defined.

Let $\varphi : \mathbb{R}^m \times Z \to \mathbb{R}$ be a continuous function, where Z is a compact space. We can define a family $(\varphi_z)_{z \in Z}$ of continuous functions, where $\varphi_z = \varphi(\cdot, z)$. Similar to the finite case we can define a "compact continuous selection" as a continuous f such that for all $x \in \mathbb{R}^m$

$$f(x) \in \{\varphi_z(x) \mid z \in Z\}$$

Similarly to the finite case, we define the set of active indices at x, $Z(x)$ as the set of all z that take the value of f at x. The following result is a useful generalization of a property of finite continuous selections.

Lemma 8. *The function $x \mapsto Z(x)$ is upper semi-continuous.*

Proof. Let $x \in \mathbb{R}^m$ and let V be an open neighbourhood of $Z(x)$. Consider for all $n \in \mathbb{N}$ the closed ball B_n of center x and radius 2^{-n}. Set

$$A_n = \{z \in Z \mid \exists y \in B_n, \varphi_z(y) = f(y)\}$$

A_n is closed : indeed if we write $g := \varphi - f$, we can write A_n as $\mathrm{proj}_Z(g^{-1}(0) \cap (B_n \times Z))$, and $g^{-1}(0) \cap (B_n \times Z)$ is compact so we get what we want.

Note that the intersection $\bigcap_n (A_n \backslash V)$ is empty : if there was a point z in such intersection, there would be a sequence $x_k \to x$ along which $\varphi_z(x_k) = f(x_k)$, so $z \in Z(x) \subset V$. By compactness, and because (A_n) is decreasing, there is some n such that $A_n \subset V$. Let $y \in B_n$ for such n. We have

$$Z(y) \subset A_n \subset V$$

so we get the desired result. □

The following theorem, which is a slightly modified version of Danskin's theorem, will allow us to compute explicitly subdifferentials in a particular case.

Theorem 6 (Danskin's [3]). *Assume that for all z, φ_z is C^1, $z \mapsto d_x\varphi_z$ is continuous and $f = \max_{z \in Z} \varphi_z$.*

1. *f is convex and for all $x, v \in \mathbb{R}^m$*

$$D_x f(v) = \max_{z \in Z(x)} d_x\varphi_z(v)$$

2. *f is locally Lipschitz and we have*

$$\partial_x f = \operatorname{conv}\{d_x\varphi_z(v) \mid z \in Z(x)\}$$

Proof. We'll need 1. for 2.

1. The convexity of f is a classical result from convex analysis. Notice that for $t > 0$

$$\frac{f(x+tv) - f(x)}{t} \geq \frac{\varphi_z(x+tv) - \varphi_z(x)}{t}$$

for $z \in Z(x)$, so we get $D_x f(v) \geq d_x\varphi_z(v)$. For the other inequality, consider a positive sequence $t_n \to 0$. For all n, select $z_n \in Z(x + t_n v)$, by compactness assume it converges to $z^* \in Z$. By maximality of f, we have $\varphi_{z^*}(x) \geq \varphi_z(x), z \in Z$, so $z^* \in Z(x)$. By convexity of f and φ_{z_n}, we get

$$\begin{aligned}
D_x f(v) &\leq \frac{f(x+t_n v) - f(x)}{t_n} \\
&= \frac{\varphi_{z_n}(x+t_n v) - \varphi_z^*(x)}{t_n} \\
&\leq \frac{\varphi_{z_n}(x+t_n v) - \varphi_{z_n}((x+t_n v) - t_n v)}{t_n} \\
&\leq d_x\varphi_{z_n}(v) \\
&\to d_x\varphi_{z^*}(v)
\end{aligned}$$

so $D_x f(v)$ is indeed the maximum.

2. All φ_z are locally Lipschitz. By compactness of Z, we can find a Lipschitz constant for f.

We now show the equality involving ∂f. For that we use the results of the discussion in section 1.1.2. For the right to left inclusion, it suffices to observe that in virtue of 1., for $z \in Z(x)$

$$\forall v \in \mathbb{R}^m, f(x+v) \geq f(x) + d_x\varphi_z(v)$$

Assume that there is some $l \in \partial_x f$ not in the convex hull of $d_x\varphi_z$, $z \in Z(x)$. Because this convex hull is compact, we choose a strictly separating hyperplane : we choose v and c such that

$$\langle l, v\rangle > c > \langle d_x\varphi_z, v\rangle, z \in Z(x)$$

This contradicts the fact that $D_x f(v) = \max_{h\in\partial_x f}\langle h, v\rangle$, because $D_x f(v) = \max_{z\in Z(x)}\langle d_x\varphi_z, v\rangle$

□

We are now ready to compute the subdifferential of $\operatorname{dist}_{Y|X}$.

Theorem 7. *Let $x \in X\backslash Y$. We have*

$$\partial_x(\operatorname{dist}_{Y|X}) = \operatorname{proj}_{T_xX}\left(\operatorname{conv}\left\{\frac{x-y}{|x-y|} \mid y \in Y(x)\right\}\right)$$

where $\operatorname{proj}_{T_xX}$ *is the orthogonal projection onto T_xX.*

Proof. We first compute the subdifferential in the case where $X = \mathbb{R}^m$, and we deduce the general case. Let $\eta_Y = \frac{1}{2}\operatorname{dist}^2_Y$. We write for $x \in \mathbb{R}^m\backslash Y$

$$\eta_Y(x) = \frac{1}{2}|x|^2 - \left(\max_{y\in Y}\langle x, y\rangle - \frac{1}{2}|y|^2\right)$$

The function $\varphi_y(x) = \langle x, y\rangle - \frac{1}{2}|y|^2$ is convex. If we denote by $h(x)$ the term in parenthesis, theorem 6 yields

$$\partial_x h = \operatorname{conv}\{\nabla_x\varphi_y \mid y \in Y(x)\} = \operatorname{conv}(Y(x))$$

We can apply the theorem because in a small neighbourhood of x, the active indices stay confined to a compact set.
By using that $\eta_Y = \frac{1}{2}|\cdot| - h$, we get

$$\partial_x \eta_Y = \{x - y \mid y \in Y(x)\}$$

If we now write $\operatorname{dist}_Y = \sqrt{2\eta_Y}$, and use the subdifferential chain rule applied to the diffeomorphism $\sqrt{\cdot}$ (because $\eta_Y(x) \neq 0$, $x \notin Y$) we get

$$\begin{aligned}\partial_x \operatorname{dist}_Y &= \frac{1}{\sqrt{2\eta_Y(x)}} \operatorname{conv}\{x - y \mid y \in Y(x)\} \\ &= \operatorname{conv}\left\{ \frac{x-y}{|x-y|} \mid y \in Y(x) \right\}\end{aligned}$$

because $\eta_Y(x) = \frac{1}{2}|y - x|^2$ for all $y \in Y(x)$.

Denote $g = \operatorname{dist}_Y$, and $f = \operatorname{dist}_{Y|X}$. We will show that $\partial_x f = \operatorname{proj}_{T_x X}(\partial_x g)$.

We start from left to right. We want to show that $x \mapsto \operatorname{proj}_{T_x X}(\partial_x \operatorname{dist}_Y)$ is upper-semi continuous (where the set is seen in TX), and contains $\nabla_x f$ whenever defined. The first part is easy by semi-continuity of the original function. For the second, we assume $\nabla_x f$ is defined for some $x \in X$. By the computations made earlier and Danskin's theorem, we write

$$D_x g(v) = \min_{y \in Y(x)} \langle \nabla_x g_y, v \rangle$$

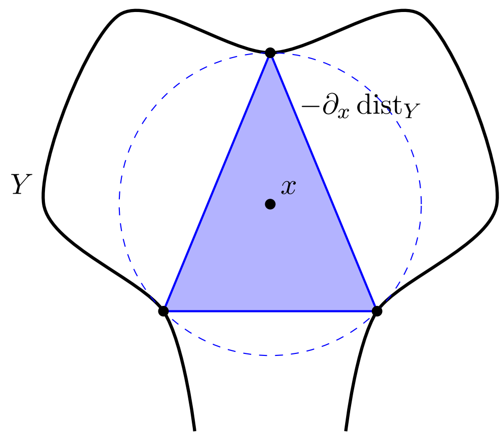
where $g_y(x) = |x - y|$. Note that because $\nabla_x f$ exists, this is linear if restricted to $T_x X$. By using linearity, we find that there is $y \in Y(x)$ such that for all $v \in T_x X$

$$D_x g(v) = \langle \nabla_x g_y, v \rangle = \langle \nabla_x f, v \rangle$$

Hence $\nabla_x g_y$ projects onto $\nabla_x f$, so we have what we want. The inclusion then follows from minimality of subdifferentials.

For the inclusion from right to left, we consider $x_k \to x \in X$ a sequence such that $\nabla_{x_k} g$ is well defined and converges, and we will show that it is a subgradient of $\operatorname{dist}_{Y|X}$ up to projection. We consider up to restricting to a neighbourhood of x, that $X = \mathbb{R}^n$ where $n = \dim(X)$. Let $(V_k)_k$ be a decrasing sequence of open neighbourhoods of $\partial_x f$ converging to it. By semi-continuity of ∂g and up to extracting, $\nabla_{x_k} g \in V_k \times \mathbb{R}^{m-n}$, So the projection of $\nabla_{x_k} g$ converges to a point in $\partial_x f$, we got equality. $\square$

Figure 3: Illustration of theorem 7



3.2 Special points of the distance function

We now consider $X \subset \mathbb{R}^m$ to be some non specific submanifold. We denote in this section $g : (a, b) \mapsto |a - b|^2$ and $g_a : x \mapsto |a - x|^2$ for all $a \in \mathbb{R}^m$.
Following [2], we define

Definition 5 (Medial axis). *We define M_Y, the medial axis of Y as*

$$M_Y = \{x \in \mathbb{R}^m \mid |Y(x)| \geq 2\}$$

Some other special points of the distance function include the cut-locus of Y, that we'll denote $\mathrm{CLoc}(Y)$ and the central set of Y that we denote $C(Y)$, introduced respectively by Thom [15] and Milman [6].

Definition 6 (Central set). *An open ball $B \subset \mathbb{R}^m \backslash Y$ will be called maximal if there is no open ball in $\mathbb{R}^m \backslash Y$ strictly containing B. The central set $C(Y)$ is then the set of the centers of all maximal balls.*

If Y is smooth we can also define the following set

Definition 7 (Cut-locus). *The cut-locus of Y is defined as follows*

$$\mathrm{CLoc}(Y) = M_Y \cup \{x \in \mathbb{R}^n \mid g_{x|Y} \textit{ has a degenerate global minimum}\}$$

If we assume that Y is a submanifold of $\mathbb{R}^m$, we can link these seemingly differents concepts by the following proposition, generalizing the arguments from [6].

Proposition 5. *Assume Y is a smooth submanifold of $\mathbb{R}^m$. Then $\overline{M_Y} = \mathrm{CLoc}(Y) = C(Y)$*

Proof. We first show $\overline{M_Y} \subset \mathrm{CLoc}(Y)$. Consider a sequence $x_k \to x$, where $x_k \in M_Y$ and x is not. Let y be the single element in $Y(x)$, and for all k, $y_k, y_k' \in Y(x_k)$ two different minimizers. Then for all k, $\nabla_{y_k} g_{x_k} - \nabla_{y_k'} g_{x_k} = 0$ and we can show that $\mathrm{Hess}_y(g_{x|Y})$ vanishes in some direction $u \in T_y Y$, defined as an extracted limit of $\frac{y_k - y_k'}{|y_k - y_k'|}$. Hence $x \in \mathrm{CLoc}(Y)$.

We now show $\mathrm{CLoc}(Y) \subset C(Y)$. We have $M_Y \subset C(Y)$: indeed if for $x \in M_Y$ there is a ball B' containing $B := B(x, \mathrm{dist}_Y(x))$, then $\partial B'$ and ∂B must intersect tangentially at $Y(x)$. This set has at least two points so we find that the center of B' must be x, so $x \in C(Y)$.

Let $x \in \mathrm{CLoc}(Y)$ such that $x \notin M_Y$. So $g_{x|Y}$ has a degenerate global minimum. Let y be this minimum, $R = |x - y|$ and denote

$$\nu = \frac{x - y}{R} \in N_y Y$$

Minimality gives us $\mathrm{Hess}_y(g_{x|Y}) \succeq 0$, hence if we denote by κ_i the principal curvatures in direction ν of Y i.e the eigenvalues of the second fondamental form in direction ν at y, then $\kappa_i \leq \frac{1}{R}$. By [12], $\mathrm{Hess}_y(g_{x|Y})$ being degenerate implies that for some i, $\kappa_i = \frac{1}{R}$.

If we now assume that there is a strictly larger maximal ball B' containing $B(x, R)$, B' must also be tangent to $\partial B(x, R)$ at y. The center of B' then lies on a line with y and x. Denote $R' > R$ the radius of B'. Applying the same arguments to B', $\kappa_j \leq \frac{1}{R'}$ for all j. But for some i, $\kappa_i = \frac{1}{R} > \frac{1}{R'}$, contradiction. So $B(x, R)$ is a maximal ball, $x \in C(Y)$.

Lastly, we show that $C(Y) \subset \overline{M_Y}$. Assume for contradiction that there is some $x \in C(Y)$ not in $\overline{M_Y}$. We find $\varepsilon > 0$ such that for $z \in B := B(x, \varepsilon)$, $Y(z)$ is a singleton. We can define $\eta : B \to Y$ that to a point z associates the unique point in $Y(z)$. This is a continuous function by semi continuity of $Y(z)$.

Denote $z_0, z_1 \in \partial B$ respectively the closest and farthest intersection of the line passing through y and x. Observe that because $x \in C(Y)$, $\eta(z_1) \neq y$ and $\eta(z_0) = y$. Up to shrinking B, we find a neighbourhood $V \subset Y$ of y containing $\eta(B)$, and a section $s : V \to NY$ such that for all $y' \in V$, the line passing through y' in direction $s(y')$ goes through ∂B, and in particular, the one at y passes through z_0 and z_1. We denote by $S \subset \partial B$ the set of these intersection points. From now on every topological operation happens in S.

Note that by our assumptions, η is a homeomorphism on the hemisphere of S containing z_0 over its image. We denote the open hemisphere U. We now show that there is a point $z' \in S\backslash U$ such that $\eta(z') = y$. This will conclude the proof, as the only normal line to y intersecting S goes through z_0 and z_1, so $z' = z_1$ and this is a contradiction.

Assume there is no such point z'. Because $y \in \eta(U)$ is not in the image of $\eta_{|S\backslash U}$ and η sends the cycle ∂U to $\partial(\eta(U))$, this last cycle is not homologically trivial. But $\partial U = \partial(S\backslash U)$, so the cycle ∂U is homologically trivial in $S\backslash U$, which contradicts what we found. $\square$

We now relate these special sets of points to differentiability of dist_Y. Using similar computations as in theorem 7, we can show, because dist_Y and its square have the same differentiability points out of Y, that outside of $Y \cup M_Y$ by Danskin's theorem, dist_Y is differentiable. In fact we have the following stronger statement.

Proposition 6. *The function* dist_Y *is* C^1 *on* $\mathbb{R}^m\backslash(Y \cup \overline{M_Y})$.

Proof. Let $x \in \mathbb{R}^m\backslash(Y\cup\overline{M_Y})$. Let U be a neighbourhood of x contained in $Y\cup\overline{M_Y}$. Similarly to proposition 5 we can then define a continuous function $\eta : U \to Y$ that to each $z \in U$ associates the unique $y \in Y(z)$. dist_Y is differentiable on U. Using the first part of Danskin's theorem and similar computations to theorem 7, we get

$$d_z \operatorname{dist}_Y = d_z g_{\eta(z)}$$

From this, we deduce that $d \operatorname{dist}_Y$ is continuous on U. $\square$

3.3 Reducing to local continuous selections

In this section, $Y \subset \mathbb{R}^m$ will be a smooth submanifold, as X. Our goal will be to show that under certain assumptions, the distance function is locally a continuous selection of finitely many C^2 functions, and so enters the Morse theory framework developped in section 2.

We'll denote by φ the Riemannian exponential map restricted to NY, i.e for $(y, v) \in NY$, $\varphi(y, v) = y + v$. The following proposition from [2] uses this map to find points where dist_Y is locally a continuous selection.

Proposition 7. *Assume Y is smooth and let $x \in \mathbb{R}^m$ be a regular value of φ. Then $Y(x) = \{y_1, ..., y_n\}$ is finite. Moreover there exists*

- *V a neighbourhood of x*
- *For all i, W_i a neighbourhood of y_i such that $\overline{W_i}$ are disjoint*

that verify

$$\forall z \in V, \operatorname{dist}_Y(z) = \min_i \operatorname{dist}_{Y\cap\overline{W_i}}(z)$$

and all $\operatorname{dist}_{Y\cap\overline{W_i}}$ *are smooth on V.*

Proof. We first prove that $Y(x)$ is finite. This set is compact. If we write $\pi : NY \to Y$ the natural projection, we get $Y(x) \subset \pi(\varphi^{-1}(x))$. But note that x being a regular value, $\varphi^{-1}(x)$ is of codimension m in NY, hence it is a set of disjoint points. The same applies after applying π, so $Y(x)$ is finite.

We now construct the neighbourhoods : denote $Y(x) = \{y_1, ..., y_n\}$. For all i, because x is regular, we choose a neighbourhood U_i of $x - y_i$ in $\mathbb{R}^{m-\dim(Y)} \cong N_{y_i}Y$, W_i a neighbourhood of y_i in $\mathbb{R}^m$, and V_i a neighbourhood of x in $\mathbb{R}^m$ such that, up to trivialization, $(W_i \cap Y) \times U_i \subset NY$, and $\varphi : (W_i \cap Y) \times U_i \to V_i$ is a diffeomorphism.

Now note that for all z in some neighbourhood $V \subset \bigcap_i V_i$ of x, $z - y$ is close to $x - y_i$ for all $y \in W_i \cap Y$. So if it happens that $z - y \in N_y Y$, then $z - y \in U_i$. From that observation, and the induced diffeomorphism, we deduce that there is only one such y. From that we get

$$\forall z \in V, \operatorname{dist}_{Y\cap\overline{W_i}}(z) = |\pi(\varphi_i^{-1}(z)) - z|$$

where $\varphi_i := \varphi_{|(W_i \cap Y)\times U_i}$. So the distance is indeed a smooth function on V.

Lastly, by semi-continuity of $x \mapsto Y(x)$, we can assume that V is small enough such that for $z \in V$, $Y(z) \subset \bigcup_i W_i$. Hence

$$\mathrm{dist}_Y(z) = \min_i \mathrm{dist}_{Y\cap \overline{W_i}}(z)$$

□

A natural question following this result would be to consider how restrictive is the condition of being a regular value of φ, i.e what are these points, is it likely that a critical point of $\mathrm{dist}_{Y|X}$ is not a regular value and what can we say in this case.

We can first observe that critical values of φ are points in $\mathbb{R}^m$ verifying a degeneracy condition similar to the one defining the cut-locus $\mathrm{CLoc}(Y)$: x is a critical value if there is $y \in Y(x)$ that is a degenerate minimum of the function $y' \in Y \mapsto |x - y'|$. These points do not match $\mathrm{CLoc}(Y)$ though, as a regular value x of φ can be such that $Y(x)$ is not a singleton.

If we try to generalize the statement by dropping the regular value condition, we can still show using the semi-continuity of the active index and compactness that there is finitely many neighbourhoods W_i such that for z close enough to x, $\mathrm{dist}_Y(z)$ can still be obtained by minimizing over $\mathrm{dist}_{Y\cap\overline{W_i}}(z)$.
We do not get smoothness of these distances near x for arbitrary Y, a counterexample is given by $Y = \mathbb{S}^{m-1}$ and $x = 0$: if for some choice of W_i, the distances were smooth, then $\partial_0(\mathrm{dist}_{\mathbb{S}^{m-1}\cap\overline{W_i}})$ would consist of a single point, which is not the case by theorem 7.

On the "likelyhood" side, it is shown that when Y is algebraic, then in the generic case, critical points of the distance do not intersect critical values of φ [2].

4 Algebraic degrees and critical points

In this section, we still place ourselves in the euclidean space $\mathbb{R}^m$. We will consider algebraic submanifolds $X, Y \subset \mathbb{R}^m$ defined as zero sets of polynomials, and study critical points of functions over these. More precisely, we will focus on critical points of linear functions, and of $\mathrm{dist}_{Y|X}$, and try to give an "algebraic count" of these, upper bounding the number of real critical points of these functions.

4.1 Constructions from metric algebraic geometry

To give an algebraic count of critical points, that are defined through real polynomial equations, the standard thing to do is to complexify the set of equations i.e look for complex solutions of the real system of polynomials. A second step will be to "compactify" the set where the solutions live, which will be done by an embedding in complex projective space. Lastly, after the projectification step, we will introduce the tools we will need from intersection theory.

4.1.1 Complexification

Let $X \subset \mathbb{R}^m$ be an algebraic manifold defined as the zero set of the follows polynomials : $p_1, ..., p_r$. If we view $p_1, ..., p_r \in \mathbb{R}[x_1, ..., x_m] \subset \mathbb{C}[x_1, ..., x_m]$ as complex polynomials, we define $X_\mathbb{C}$ as the submanifold of $\mathbb{C}^m$ defined as the complex zero set of $p_1, ...p_r$.
We also need to define what it means for some complex vector to be tangent or normal $X_\mathbb{C}$. We define the tangent bundle of $X_\mathbb{C}$ as follows :

$$T(X_\mathbb{C}) = \left\{ (x, v) \mid x \in X, v \in \ker \begin{pmatrix} \nabla_x p_1 \\ \vdots \\ \nabla_x p_r \end{pmatrix} \right\} \tag{4}$$

and the normal bundle

$$N(X_\mathbb{C}) = \{v \in \mathbb{C}^m \mid \forall w \in T_x X_\mathbb{C}, \sum_{i=1}^m v_i w_i = 0\} \tag{5}$$

Note that the product involved is not an inner product on $\mathbb{C}^m$. In particular, one has $N_x(X_\mathbb{C}) \cap T_x(X_\mathbb{C}) \neq 0$ for all $x \in X_\mathbb{C}$, as there is a whole $m-1$ dimensional quadric called the *isotopic quadric* where this product vanishes :

$$Q = \{(x_1, ..., x_m) \in \mathbb{C}^m \mid \sum_{i=1}^m x_i^2 = 0\}$$

Using this product to define $N(X_\mathbb{C})$ is the approach commonly followed in metric algebraic geometry for two reasons:

1. A hermitian inner product such as $(x, y) \mapsto \bar{x}_1 y_1 + \cdots + \bar{x}_m y_m$ is not algebraic over $\mathbb{C}$, which would undermine the usage of algebraic geometry for these problems;

2. Even after complexification, in applications we are usually only interested in the real solutions, which are still captured by this definition.

This algebraic product has the added benefit that it generalizes the following identity

$$N_x(X_\mathbb{C}) = \mathrm{span}\{\nabla_x p_1, ... \nabla_x p_r\}$$

for all $x \in X$. For what follows, we omit the subscript and directly write $X, Y \subset \mathbb{C}^m$, considering them as already complexified.

4.1.2 Tangency and normality in projective space

Although complexification unlocks new tools for studying equations (ACP), complex affine space is not ideal, one of the reasons being the fact it is not compact. A classical thing to do in algebraic geometry is to compactify the affine space, viewing it as embedded in a projective space, where a lot of efficient tools have been developped.

Our goal will be then to embed our spaces of interest, $X, Y \subset \mathbb{C}^m$ algebraic manifolds and the related bundles, inside a complex projective space. Embedding X, Y is easy : consider the canonical embedding $\mathbb{C}^m \hookrightarrow \mathbb{P}^m$, that first embeds $\mathbb{C}^m$ in $\mathbb{C}^m \times \{1\}$ then projects it into projective space. We embed X and Y in their closure $\bar{X}$, $\bar{Y}$ inside $\mathbb{P}^m$.

The troublesome part is to construct natural tangent and normal bundles over $\bar{X}$ and $\bar{Y}$ from those in affine space. Define the affine tangent and normal bundles over X (and respectively Y) as the affine bundles of fiber over $x \in X$

$$\hat{T}_x X = x + T_x X \text{ and } \hat{N}_x X = x + N_x X \tag{6}$$

These are our bundles of interest. When taking the projective closure $\overline{\hat{T}X}$, $\overline{\hat{N}X}$ of the affine tangent and normal bundles, we would like there to exist vector bundles $\hat{T}\bar{X}$, $\hat{N}\bar{X}$ over $\bar{X}$ such that

$$\overline{\hat{T}X} = \mathbb{P}(\hat{T}\bar{X}), \quad \overline{\hat{N}X} = \mathbb{P}(\hat{N}\bar{X}).$$

where $\mathbb{P}(\hat{T}\bar{X})$ is the image of $\hat{T}\bar{X}$ through the usal projection onto projective space (and similarly for $\hat{N}\bar{X}$). Calling $\hat{T}\bar{X}$ the embedded tangend bundle and $\hat{N}\bar{X}$ the euclidean normal bundle, we would like moreover to have a definition of these bundles for any projective variety, independently of whether they come from an affine variety or not.

Suppose we are given a nonsingular projective variety

$$X = \{x \in \mathbb{P}^m \mid p_1(x) = \cdots = p_r(x) = 0\} \subset \mathbb{P}^m$$

where $p_1, \ldots, p_r \in \mathbb{C}[x_1, \ldots, x_m]$ are homogeneous polynomials, then the only natural way to define the embedded tangent bundle $\hat{T}X \subset X \times \mathbb{P}^m$ is through the same equations as (4).
On the other hand, for the Euclidean normal bundle there are at least two options:

1. Fix a non-degenerate quadric $Q \subset \mathbb{P}^m$. We can identify the projective variety $X \subset \mathbb{P}^m$ with the affine cone $C(X) \subset \mathbb{C}^{m+1}$, and using the definition of tangent space of an affine variety we have $\hat{T}^c_{[x]} X = T_x C(X)$ and can define
$$\hat{N}^c_{[x]} X := \mathbb{C}x + (T_x C(X))^{\perp}$$
for any $x \in \mathbb{C}^{m+1} \setminus 0$, where $^\perp$ is the Q_∞-orthogonal in $\mathbb{C}^{m+1}$. This is the same as defining $\hat{N}X$ through the same equations as (5) and adding the subspace $\langle x \rangle$.

2. Fix a hyperplane $H_\infty \subset \mathbb{P}^m$ and a non-degenerate quadric $Q_\infty \subset H_\infty$. Starting from a tangent plane $\mathbb{P}(\hat{T}_x X)$ meeting H_∞ transversely, we can define
$$\mathbb{P}(\hat{N}_x X) := \mathbb{C}x + (\mathbb{P}(\hat{T}_x X) \cap H_\infty)^{\perp}$$
where $^\perp$ is the Q-orthogonal in H_∞. When
$$H_\infty = \mathbb{P}^m \setminus \mathbb{C}^m = \{x \in \mathbb{P}^m \mid x_{m+1} = 0\}, \quad Q_\infty = \{x \in H_\infty \mid x_1^2 + \cdots + x_m^2 = 0\}$$
and $Z \subset \mathbb{C}^m$ is an affine variety such that $X = \bar{Z}$, then this definition of $\mathbb{P}(\hat{N}_x X)$ agrees with $\overline{\hat{N}_x Z}$.

In general $\hat{N}_x X \neq \hat{N}^c_x X$, but under the right hypotheses they both form bundles $\hat{N}X$, $\hat{N}^c X$ which are isomorphic. To distinguish them, we will call $\hat{N}^c X$ the cone Euclidean normal bundle.

The first intepretation is used in [8, 7] and leads to beautiful duality theorems. The second interpretation is defined and used in [5, 14] and is more natural when considering projective varieties constructed as projective closures of affine varieties. In the following sections, we will use this second interpretation, as we started with affine X and Y.

Embedded tangent bundle Let V be an $(m+1)$-dimensional vector space over $\mathbb{C}$, and denote by $\mathbb{P}^m := \mathbb{P}(V)$. Let $X \subset \mathbb{P}^m$ be a nonsingular variety of dimension n and codimension $c := m - n$.
We first define the **embedded tangent bundle** as follows. Consider the Euler sequence

$$0 \to \mathcal{O}_{\mathbb{P}^m}(-1) \to V_{\mathbb{P}^m} \to \mathcal{T}_{\mathbb{P}^m}(-1) \to 0$$

where $V_{\mathbb{P}^m} := \mathbb{P}^m \times V$. Each of these terms has a clear geometric meaning: for any $x \in \mathbb{P}^n$ the fibers are

$$0 \to \mathbb{C}x \to V \to V/\mathbb{C}x \to 0.$$

The restriction of the Euler sequence yields

$$0 \to \mathcal{O}_X(-1) \to V_X \to \mathcal{T}_{\mathbb{P}^m}(-1)|_X \to 0$$

where $V_X := X \times V$, and the fibers look the same as before. Consider the tangent bundle $\mathcal{T}_X$ and normal bundle $\mathcal{N}_X$ in the exact sequence

$$0 \to \mathcal{T}_X(-1) \to \mathcal{T}_{\mathbb{P}^m}(-1)|_X \to \mathcal{N}_X(-1) \to 0$$

with fibers at $x \in X$

$$0 \to T_xX \to V/\mathbb{C}x \to \frac{V/\mathbb{C}x}{T_xX} \to 0.$$

By considering the pullback of the inclusion $\mathcal{T}_X(-1) \hookrightarrow \mathcal{T}_{\mathbb{P}^m}(-1)|_X$ through the quotient $V_X \twoheadrightarrow \mathcal{T}_{\mathbb{P}^m}(-1)|_X$ we obtain the *embedded tangent bundle* $\hat{T}X$ fitting the following exact diagram:

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{O}_X(-1) \arrow[r] \arrow[d, equal] & \hat{T}X \arrow[r] \arrow[d] & \mathcal{T}_X(-1) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{O}_X(-1) \arrow[r] & V_X \arrow[r] \arrow[d] & \mathcal{T}_{\mathbb{P}^m}(-1)|_X \arrow[r] \arrow[d] & 0 \\
 & & \mathcal{N}_X(-1) \arrow[r, equal] \arrow[d] & \mathcal{N}_X(-1) \arrow[d] & \\
 & & 0 & 0 &
\end{tikzcd} \tag{7}$$

By definition the fiber is

$$\hat{T}_xX = (V \twoheadrightarrow V/\mathbb{C}x)^{-1}(T_xX) \subset V.$$

Its projectivization $\mathbb{P}(\hat{T}_xX) \subset \mathbb{P}^m$ is what we geometrically consider to be the tangent space at x of X: if $X = \{x \in \mathbb{P}^m \mid p_1(x) = \cdots = p_r(x) = 0\}$ then

$$\hat{T}_xX = \ker \begin{pmatrix} \nabla_x p_1 \\ \vdots \\ \nabla_x p_r \end{pmatrix}. \tag{8}$$

The bundle $\hat{T}X$ has rank $n+1$. We observe [11][(1)] that $\hat{T}X$ fits the same exact sequence (and thus is isomorphic to) the dual of the first jet bundle $J^1(\mathcal{O}_X(1))$.

Euclidean normal bundle We first need to fix a hyperplane $V' \subset V$ and a non-degenerate bilinear form B on V', which induces an isomorphism $L_{V'} \colon V' \xrightarrow{\sim} (V')^*$. These define the *hyperplane at infinity*

$$H_\infty := \mathbb{P}(V') \subset \mathbb{P}^m$$

and the *isotropic quadric at infinity*

$$Q_\infty := \{x \in H_\infty \mid B(x,x) = 0\}.$$

Geometrically it is equivalent to directly fix (H_∞, Q_∞) which is the same as fixing B up to a non-zero scalar factor. But it is handy to have the bilinear form available for what we are about to do. Set $X_\infty := X \cap H_\infty$. Consider the composition

$$\mathcal{N}_X^*(1) \hookrightarrow V_X^* \twoheadrightarrow (V')_X^* \xrightarrow{\sim} V'_X \tag{9}$$

where the first arrow comes from taking duals in (7), the second is the dual of the inclusion $V' \subset V$ and the third is $L_{V'}^{-1}$. Fiberwise for any $x \in X$ the composition looks like

$$\mathrm{Ann}(\hat{T}_x X) \hookrightarrow V^* \twoheadrightarrow (V')^* \xrightarrow{\sim} V'$$

and fails to be injective precisely when the second map is not injective. Since its kernel is $\mathrm{Ann}(V')$, this happens if and only if

$$\begin{aligned}\mathrm{Ann}(\hat{T}_x X) \cap \mathrm{Ann}(V') \neq 0 &\Leftrightarrow \hat{T}_x X + V' \neq V \\ &\Leftrightarrow \hat{T}_x X \subset V' \\ &\Leftrightarrow X \not\pitchfork_x H_\infty.\end{aligned}$$

Thus, the composition (9) is an embedding if and only if $X \pitchfork H_\infty$. In this case, the image is a sub-bundle of $(V')_X$ of rank c which we call the *bundle of normal directions* $N^\infty X$. By definition, $N^\infty X \cong \mathcal{N}_X^*(1)$. For any $x \in X$, by definition the fiber is

$$N_x^\infty X = \{v' \in V' \mid B(v, \hat{T}_x X \cap V') = 0\} = (\hat{T}_x X \cap V')^\perp \tag{10}$$

Another case which will be of interest to us is when the composition (9) is not an embedding, but the image is still a sub-bundle of $(V')_X$: this is the case when $\hat{T}_x X \subset V'$ for all $x \in X$, which is equivalent to the condition $X \subset H_\infty$. In this situation, we still denote the image by $N^\infty X$, which is a sub-bundle of $(V')_X$ of rank $c - 1$.

Next, consider the natural map of bundles

$$\begin{aligned}\mathcal{O}_X(-1) \oplus N^\infty X &\to V_X \\ (v, w) &\mapsto v + w.\end{aligned} \tag{11}$$

If this is an embedding (i.e. $x \notin N_x^\infty X$) we call its image the *Euclidean normal bundle* $\hat{N}X$, which is then a sub-bundle of V_X of rank $c+1$ isomorphic to $\mathcal{O}_X(-1) \oplus \mathcal{N}_X^*(1)$ (or of rank c if $X \subset H_\infty$). The fiber at $x \in X$ is given by

$$\hat{N}_x X = \mathbb{C}x + N_x^\infty X = \mathbb{C}x + (\hat{T}_x X \cap V')^\perp.$$

To give a geometric interpretation of when (11) is actually an embedding, notice first that for any $x \in Q_\infty$, $\hat{T}_x Q_\infty = \{y \in H_\infty \mid B(x,y) = 0\}$.

We show that $x \notin N_x^\infty X$ if and only $X_\infty \pitchfork_x Q_\infty$: suppose that $x \notin Q_\infty$, then $B(x,x) \neq 0$, and since $x \in \hat{T}_x X_\infty = \hat{T}_x X \cap V'$, we have $x \notin N_x^\infty X$. Suppose now that $x \in X \cap Q_\infty$. Then

$$\begin{aligned}\hat{T}_x X \pitchfork_x \hat{T}_x Q_\infty &\Leftrightarrow \hat{T}_x X \cap V' \notin \hat{T}_x Q_\infty \\ &\Leftrightarrow \exists v \in \hat{T}_x X \cap V',\ Q(x,v) \neq 0 \\ &\Leftrightarrow x \notin \{v' \in V' \mid Q(v', w) = 0\ \forall w \in \hat{T}_x X \cap V'\} \\ &\Leftrightarrow x \notin N_x^\infty X.\end{aligned}$$

Thus, when $X \not\subset H_\infty$ the conditions for $\hat{N}X$ to be a bundle are the following:

Definition 8. *A variety $X \subset \mathbb{P}^m$ is (H_∞, Q_∞)-orthogonally general [5] if*

1. *X is nonsingular;*
2. *$X \pitchfork H_\infty$;*
3. *$X_\infty \pitchfork Q_\infty$, where $X_\infty = X \cap H_\infty$.*

Cone Euclidean normal bundle Let V be an $(m+1)$-dimensional vector space over $\mathbb{C}$, and denote by $\mathbb{P}^m := \mathbb{P}(V)$. Fix a non-degenerate symmetric bilinear form B on V, defining the *isotropic quadric*

$$Q := \{x \in \mathbb{P}^m \mid B(x,x) = 0\}.$$

Then, the composition

$$\mathcal{N}_X^*(1) \hookrightarrow V_X^* \xrightarrow{\sim} V_X$$

is an embedding, whose image $N^c X$ is a sub-bundle of V_X of rank c. Denote $L_V : V_X \xrightarrow{\sim} V_X^*$ sending $(x,v) \in V_X$ to $(x, B(v,\cdot))$. For any $x \in X$, $\mathcal{N}_{X,x}^*(1) = \mathrm{Ann}(\hat{T}_x X)$, hence the fiber at $x \in X$ is described by

$$N_x^c X = L_V^{-1}(\mathrm{Ann}(\hat{T}_x X))) = (\hat{T}_x X)^{\perp}$$

The natural map of bundles

$$\begin{aligned} \mathcal{O}_X(-1) \oplus N^c X &\to V_X \\ (v,w) &\mapsto v+w. \end{aligned} \tag{12}$$

is injective at $x \in X$ if and only if $B(x,\cdot) \notin \mathrm{Ann}(\hat{T}_x X)$, which is the same as $X \pitchfork_x Q$. In fact, if we consider V as a hyperplane of a $(m+2)$-dimensional vector space W, and let $H_\infty = \mathbb{P}(V) \subset \mathbb{P}(W)$, we see that this is precisely the case in which $X \subset H_\infty$ discussed before.
In this case, the image of (12) is called the **cone Euclidean normal bundle** $\hat{N}^c X$, which is a sub-bundle of V_X of rank $c+1$ isomorphic to $\mathcal{O}_X(-1) \oplus \mathcal{N}_X^*(1)$.

We now fix the hyperplane at infinity H_∞ such that $Q \pitchfork H_\infty$ and let $Q_\infty = Q \cap H_\infty$. Then, in general the conditions that X is (H_∞, Q_∞)-orthogonally general and that $X \pitchfork Q$ are independent, but if both conditions are satisfied then $\hat{N}X \cong \mathcal{O}_X(-1) \oplus \mathcal{N}_X^*(1) \cong \hat{N}^c X$. Moreover, there are geometric relationships between $N^c X$ and $N^\infty X$.

Let $e_1, ..., e_{m+1} \in V$ be a basis such that $V' = \{v_{m+1} = 0\}$ in these coordinates and $e_1, ..., e_m$ form a B-orthonormal basis of V'. We can then define a B-orthogonal projection onto V' by the usual formula for $v \in V$

$$p_{V'}(v) = \sum_{i=1}^{m} B(v, e_i) e_i$$

and check that it is self adjoint. Hence $p_{V'}(N_x^c X) \subset N_x^\infty X$ by using their definitions as orthogonal spaces, for $x \in X$. The other inclusion require additionnal assumptions, more precisely that $\ker p_{V'} \cap N_x^c X = 0$.

We obtain this if we assume that V' is such that $e_1, ..., e_{m+1}$ form a B-orthonormal basis of V (we exclude purely complex behaviour), which is the case for the canonical B and V' in $\mathbb{C}^{m+1}$. In this case $N^\infty X$ is the B-orthogonal projection of $N^c X$.

4.1.3 Chow ring and useful classes

We now present some tools that will help us compute and define the algebraic degrees we will discuss. Let $X \subset \mathbb{P}^m$ be a reduced, irreducible complex projective variety. Let it be defined as the projective zero set of some polynomials, $p_1, ..., p_{m-n}$. X is then of dimension n.

Definition 9. *The group of k-cycles of X, where $k \leq n$ is the free abelian group generated by the $Y \subset X$ that are closed, k-dimensional subvarieties denoted $Z_k(X)$.*

Consider $W \subset X$ a $k+1$-dimensional subvariety of X and f a rationnal function over W. One can, for $Z \subset W$ a k-dimensional variety, define the order of vanishing of f along Z, denoted $\mathrm{ord}_Z(f)$ (cite) : intuitively it corresponds to the exponent $c \in \mathbb{Z}$ such that near a generic point x in Z, f behaves as some non zero factor times a degree c monomial. One then defines the divisor of f as

$$\mathrm{div}(f) = \sum_{\substack{Z \subset W \\ \dim(Z) = k}} \mathrm{ord}_Z(f) Z$$

If we denote $R_k(X)$ the subgroup of $Z_k(X)$ generated by the divisors of rationnal functions over some $k+1$ dimensional $W \subset X$, we get the following definition of the Chow group.

Definition 10. *The Chow group of X is defined as*

$$A(X) = \bigoplus_k A_k(X)$$

where for all k, $A_k(X) = Z_k(X)/R_k(X)$

As an example, if $X = \mathbb{P}^2$ then, $[\{x^2 + y^2 - z^2 = 0\}] = 2[\{x = 0\}]$ in $A(\mathbb{P}^2)$ because the following function, $f(x, y, z) = \frac{x^2+y^2-z^2}{x^2}$ is a rationnal function over a subvariety of $\mathbb{P}^2$ such that its divisor gives exactly the difference of these two terms.

For our purpose, wich is computing the EDD, equivalently solving an intersection problem, we will denote $A^k(X) := A_{n-k}(X)$, the classes of cycles of codimension k. We say that two subvarieties, $Y, Z \subset X$ are generically transverse if for the generic intersection smooth point, we indeed have transversality. The following proposition allows us to turn the Chow group into the Chow ring with a structure compatible with intersection problems.

Proposition 8 (Chow ring)**.** *There is a unique multiplication making $A(X)$ into a ring such that for $Y, Z \subset X$ subvarieties that are generically transverse, we have*

$$[Y] \cdot [Z] = [Y \cap Z]$$

Moreover the multiplication respects the grading of $A(X)$ i.e it sends $A^i(X) \times A^j(X)$ onto $A^{i+j}(X)$ for all i, j.

An important and motivating example is the case where $X = \mathbb{P}^m$: the Chow ring is then exactly $\mathbb{Z}[h]/(h^{n+1})$ where h is the class of a hyperplane. A subvariety $Y \subset \mathbb{P}^m$ of degree d and codimension k has the following class $[Y] = d.h^c$. We can for example retrive Bezout's theorem by multiplying the classes of two transversal curves. One can be more general and define a notion of degree for certain elements in the Chow ring.

Proposition 9 (Degree of a class)**.** *There is a unique morphism,* $\deg : A_0(X) \to \mathbb{Z}$ *such that* $\deg([\ \cdot\ ]) = 1$ *and for all $Y, Z \subset X$ subvarieties of summed codimensions n, we have*

$$|Y \cap Z| = \deg([Y][Z])$$

With the notion of the Chow ring, we can now introduce some useful classes that live in the Chow ring of some variety. The first are the Chern classes.

Consider a complex vector bundle over X, $E \to X$ of rank r where X is assumed to be smooth. We define the Chern classes of this bundle with the following proposition.

Proposition 10 (Chern classes)**.** *There is a unique way to assign a class $c(E) = c_0(E)+...+c_r(E) \in A(X)$, where $c_i(E) \in A^i(X)$ for all i, such that :*

1. *If E is a line bundle, then $c(E) = 1 + c_1(E)$ and $c_1(E) = [\mathrm{div}(\tau)]$ where τ is a rationnal section of E*
2. *Let $s_1, ..., s_{r+k-1}$ be global sections of E. Their degeneracy locus is the variety $W \subset X$ consisting of the points where all of these sections are linearly dependant. If $\mathrm{codim}(W) = k$ (i.e the expected one), then $[W] = c_k(E)$.*
3. *(Whitney's formula) If we have an exact sequence*

   $$0 \to F \to G \to H \to 0$$

   of vector bundles over X, then $c(F)c(H) = c(G)$.
4. *(Functoriality) If $f : Y \to X$ is a morphism of smooth varieties, then $f^*(c(E)) = c(f^*E)$ where f^*E is the pullback bundle of E and*

   $$f^* : \begin{array}{ccc} A(X) & \longrightarrow & A(Y) \\ [V] & \longmapsto & \left[f^{-1}(V)\right] \end{array}$$

One other class of interest are polar classes. Recall $\hat{T}X$ the embedded tangent bundle defined in (8). Also denote for $x \in X$, $\mathbb{P}(\hat{T}_x X)$ the projective embedding of $\hat{T}_x X$. Then for $L \subset \mathbb{P}^m$ a subspace, define the polar variety of X with respect to L as

$$P(X, L) = \overline{\{x \in X \mid \dim(\mathbb{P}(\hat{T}_x X) \cap L) > n - \mathrm{codim}(L)\}} \subset X$$

There is an intuitive example when L is a point and X an ellipsoid : place yourself at point L, and consider all the light rays arriving at your position. The light rays that form the "contour" of X from your perspective intersect X exactly at $P(X, L)$.

For generic L of codimension $n + 2 - k$, the class $[P(X, L)] \in A^k(X)$ is constant, and we define the k-th polar class of X as $p_k(X) = [P(X, L)]$. It turns out that polar classes are Chern classes of some bundle, more precisely $c_k(\hat{T}^*X) = p_k(X)$, where $\hat{T}^*X$ is the rank $n + 1$ vector bundle whose fibers are dual to those of $\hat{T}X$.

To see it, we write $L = \bigcap_i \ker \alpha_i$ with $\alpha_1, ..., \alpha_{n+2-k} \in (\mathbb{C}^{m+1})^*$. For all i, the function

$$\begin{aligned} X &\longrightarrow \hat{T}^*X \\ x &\longmapsto \alpha_{i|\hat{T}_x X} \end{aligned}$$

is a section of $\hat{T}^*X$. The degeneracy locus of all these sections is the set W of all $x \in X$ such that $\alpha_{i|\hat{T}_x X}$ are linearly dependant. This is equivalent to asking that $L \cap \mathbb{P}(\hat{T}X)$ has dimension greater than expected, i.e greater than $n - \mathrm{codim}(L)$. By condition 2., $c_k(\hat{T}^*X) = [W] = p_k(X)$.

Lastly, we present Segre classes and link them to the other ones. Consider again a vector bundle $E \to X$ of rank r. We construct a projective bundle on X i.e whose fibers are projective spaces of dimension $r - 1$, $\mathbb{P}(E) \to X$, with projection map p, by projectifying the fiber over each point $x \in X$. In this way, we have for all $x \in X$, $p^{-1}(x) = \mathbb{P}(E_x)$.

From this projective bundle, we can construct a new vector bundle, the tautological line bundle over $\mathbb{P}(E)$

$$\mathcal{O}_{\mathbb{P}(E)}(-1) = \{((x, L), w) \in \mathbb{P}(E) \times \mathbb{C}^{m+1} \mid w \in L\}$$

and $\mathcal{O}_{\mathbb{P}(E)}(1)$ its dual bundle.

Denote by $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$. We define the k-th Segre class of E for $k \in [\![0, r]\!]$ as

$$s_k(E) = p_*(\xi^{r-1+k}) \in A^k(X)$$

where the pushforward p_* is defined as follows : p must be a proper map. If this is the case, p_* is a linear map such that for $Y \subset X$ a subvariety, if $p(Y)$ has dimension strictly smaller than Y then $p_*[Y] = 0$, if not $p_*[Y] = d[p(Y)]$ where d is the degree of the map $p_{|Y}$.

Again, we have relations to the other classes : Segre classes are inverses of Chern classes i.e $s(E)c(E) = 1 \in A(X)$. Another important thing to note is that the bundle map p embeds $A(X)$ into $A(\mathbb{P}(E))$ through p^*.

4.2 Linear optimization degree

Let $X \subseteq \mathbb{C}^m$ be a smooth algebraic submanifold such that $\bar{X} \pitchfork H_\infty$ in $\mathbb{P}^m$. We will start with the easiest case of critical points, the problem of linear optimization on X : given a linear function $\alpha \in (\mathbb{C}^m)^*$, we will consider its critical points after restricting it to X.

We now show that the number of critical points is constant for generic choices of α. There is a unique $u \in \mathbb{C}^m$ such that for $x \in \mathbb{C}^m$

$$\alpha(x) = \sum_i u_i x_i$$

We also have that $x \in X$ is a critical point of $\alpha_{|X}$ iff

$$T_x X \subset \ker d_x \alpha = \ker \alpha$$

Recalling the exact sequences of bundles in section 4.1.2 (which are also valid when working in $\mathbb{C}^m$), one has that $(x, \alpha) \in \mathcal{N}_X^*(1) \hookrightarrow X \times (\mathbb{C}^m)^*$ and that the following diagram commutes

$$\begin{tikzcd}
\mathcal{N}^*_X(1) \arrow[r, hook] \arrow[d, "\pi_1"] \arrow[drr, "\pi_2"] & \mathbb{C}^m \times (\mathbb{C}^m)^* \cong \mathbb{C}^m \times \mathbb{C}^m \arrow[d, "\mathrm{pr}_2"] \\
X & (\mathbb{C}^m)^* \cong \mathbb{C}^m
\end{tikzcd}$$

Because $(x, \alpha) \in \mathcal{N}^*_X(1)$, there is a one to one correspondance between critical points of α on X and points in $\mathrm{pr}_2^{-1}(u) \cap \mathcal{N}^*_X(1)$. We then have two cases, if π_2 is dominant, because the $\mathcal{N}^*_X(1)$ has dimension n and the image has too, the cardinality of the generic fiber is constant. If it is not dominant, the cardinality of the generic fiber is 0, constant as well.

Thanks to this discussion, we define the linear optimization (LO) degree $\mathrm{LOdeg}(X)$ as the number of critical points of a generic linear functional $\alpha \in (\mathbb{C}^m)^*$ restricted to X.

We will now express the LO degree in terms of characteristic classes. We will proceed by taking the appropriate compactification in $\mathbb{P}^m$ of X and its bundles. We will without loss of generality choose the canonical compactification i.e induced by the canonical embedding $\mathbb{C}^m \hookrightarrow \mathbb{C}^m \times \{1\} \to \mathbb{P}^m$, where the hyperplane at infinity is given by $\mathbb{C}^m \times \{0\} \cong \mathbb{C}^m$.

The original problem can be framed as follows: given a generic vector hyperplane $H \subset \mathbb{C}^m$, we want to count for how many points $x \in X$ we have $H \supseteq T_xX$. This is the same as asking that $\bar{H} \supseteq \overline{T_xX}$ and equivalently that $\bar{H} \cap H_\infty \supseteq \overline{T_xX} \cap H_\infty$. Defining $T^\infty\bar{X} := (\hat{T}\bar{X}) \cap (\bar{X} \times \mathbb{C}^m)$ (which is a bundle of rank m by the transversality condition) so that

$$\overline{T_xX} \cap H_\infty = \overline{\hat{T}_xX} \cap H_\infty = \mathbb{P}(\hat{T}_x\bar{X}) \cap H_\infty = \mathbb{P}(T^\infty_x\bar{X})$$

we have reduced the problem to the following: given a generic linear functional $\alpha \in (\mathbb{C}^m)^*$, count for how many points $x \in X$ we have that the corresponding section $\alpha \in (T^\infty_x\bar{X})^*$ vanishes. Generically, there is no point at infinity $x \in \bar{X} \setminus X$ for which this is true, so that the number to be computed is really the degree of the zero locus of a generic section of $T^\infty\bar{X}$, which we know by Property 2. of Chern classes to be $\deg(c_n(T^\infty\bar{X}))$.

To calculate this value, notice that we have the exact sequence

$$0 \to T^\infty\bar{X} \to \hat{T}\bar{X} \to \mathbb{C}^1_{\bar{X}} \cong \mathcal{O}_{\bar{X}} \to 0$$

where fiberwise we use the isomorphism $\hat{T}_x\bar{X}/(\hat{T}_x\bar{X} \cap \mathbb{C}^m) \cong \mathbb{C}$. Then, by Property 3. of Chern classes it follows that $c(\hat{T}\bar{X}) = c(T^\infty\bar{X})c(\mathcal{O}_{\bar{X}}) = c(T^\infty\bar{X})$ and thus we conclude

$$\mathrm{LOdeg}(X) = \deg(c_n(\hat{T}\bar{X})) = \deg(p_n(\bar{X})),$$

i.e. the LO degree is the degree of the top polar class of $\bar{X}$.

Although we do not explain it here, the discussion around the LO degree can be extended to the case of singular varieties (cite sources Lorenzo + OG sources) and

4.3 Critical points of the euclidean distance

We first introduce some notation : Y is defined by the zero set of polynomials $q_1, ..., q_n$, while X is defined by the zero set of $p_1, ..., p_r$. We will assume as stated in the introduction that X and Y are smooth algebraic manifolds.

From theorem 7 proved in the last section, we now that a point $x \in X$ is critical for the function $\mathrm{dist}_{Y|X}$ iff there exists a number $k \geq 1$, $y_1, ..., y_k \in Y$ and $\lambda_1, ..., \lambda_k \in \mathbb{R}$ such that :

$$\begin{array}{ll} 1. & \text{The } y_i \text{ are all distinct} \\ 2. & \forall i, |x - y_i| = \mathrm{dist}_Y(x) \\ 3. & \sum_i \lambda_i \frac{x-y_i}{|x-y_i|} \in N_xX \\ 4. & \forall i, \lambda_i \geq 0 \text{ and } \sum_i \lambda_i = 1 \end{array} \tag{TCP}$$

We want to transform these into mostly algebraic equations, such that the solutions of these new equations include true critical points of the distance function. The following equations with unknowns $x, y_1, ..., y_k$ accomplish that

$$\begin{array}{ll} 1. & \text{The } y_i \text{ are all distinct} \\ 2. & \forall i, j, |x - y_i|^2 = |x - y_j|^2 \text{ and } x - y_i \in N_{y_i}Y \\ 3. & \text{The family } (x - y_i)_i \text{ is linearly dependent in } \mathbb{C}^m/N_xX \end{array} \tag{ACP}$$

The first condition is an open one. The second one is algebraic because $N_{y_i}Y$ is spanned by $\nabla_{y_i}q_1, ..., \nabla_{y_i}q_n$, so $x - y_i \in N_{y_i}Y$ is equivalent to asking that the family $\{x - y_i, \nabla_{y_i}q_1, ..., \nabla_{y_i}q_n\}$ is linearly dependent. The third condition is algebraic as well, as it amounts to ask

$$\mathrm{rk}(x - y_1, ..., x - y_k, \nabla_x p_1, ..., \nabla_x p_r) \leq \mathrm{rk}(\nabla_x p_1, ..., \nabla_x p_r)$$

which can be checked via polynomials.
One also checks that solutions to (TCP) are solutions of (ACP) : condition 1. stays the same, condition 2. from (TCP) implies that $x - y_i \in N_{y_i}Y$ and equalities of the distances, and condition 3. and 4. imply that $x - y_i$ are linearly dependant in $\mathbb{C}^m/N_xX$.

In what follows, we will study particular cases of these equations and connect them to already existing notions on algebraic manifolds. The case where $Y = \{y\}$ will lead to the notion of *euclidean distance degree*, and when $k = 2$ and $X = \mathbb{C}^m$ we will get the notion of *bottleneck degree.*

4.3.1 Euclidean distance degree

As stated earlier, we consider the case where $Y = \{y\}$ and X is some algebraic submanifold of $\mathbb{C}^m$. (ACP) then simplifies to

$$x - y \in N_xX$$

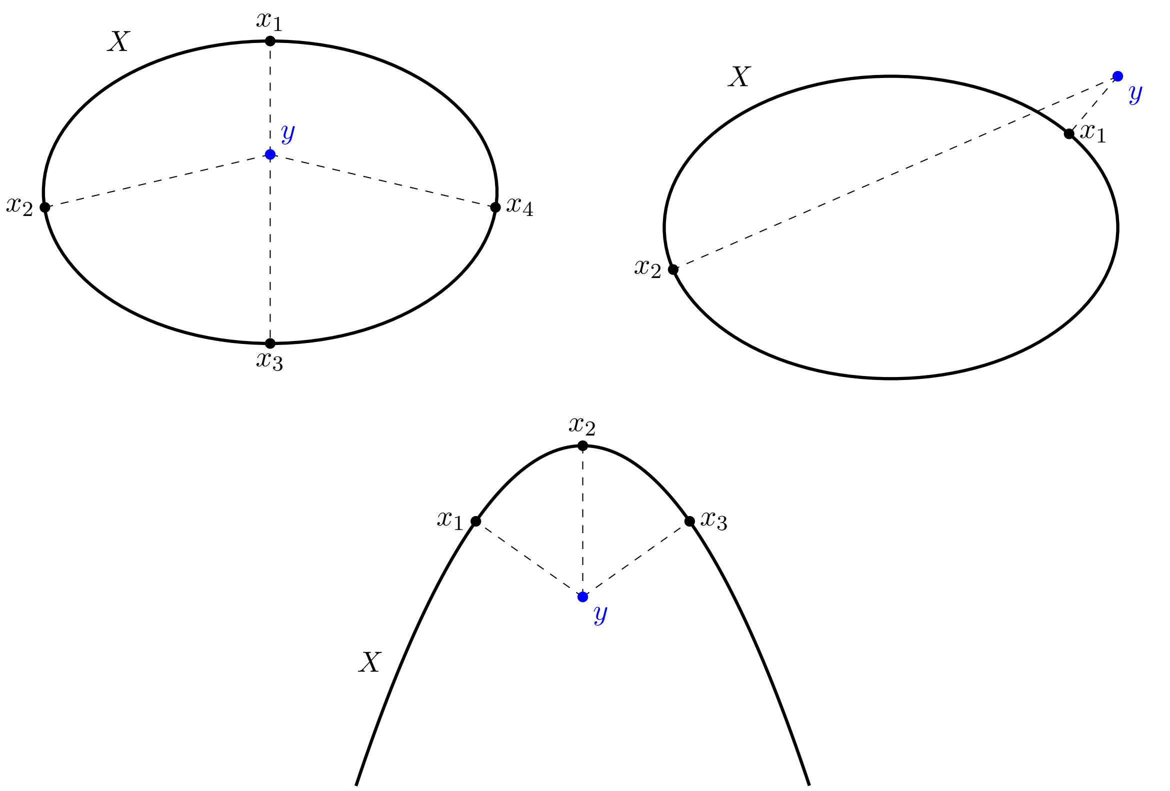
where the unknown is x. This is equivalent to asking that $y \in x + N_xX$ i.e $y \in \hat{N}_xX$ as defined in (6). This bundle has two natural projections $\pi_1 : \hat{N}X \to X$ and $\pi_2 : \hat{N}X \to \mathbb{C}^m$. The set of solutions to (ACP) is then $\pi_1(\pi_2^{-1}(y))$. Using the fact that $\hat{N}X$ is of dimension m, one can show the following property.

Proposition 11. *For generic $y \in \mathbb{C}^m$, the set $\pi_1(\pi_2^{-1}(y))$ is finite and it has a constant number of elements.*

The *euclidean distance degree* (EDD) of X is then defined as this generic number of solutions to the critical point equation. It then gives an upper bound on the number of real solutions to (ACP).

Consider the example where $X \subset \mathbb{C}^2$ is an ellipse. It is then defined by a degree 2 polynomial and we will show later that the euclidean distance degree of X is 4. We see that for generic points y in the bounded region of the plane $\mathbb{R}^2$ created by X, we get the maximum amount, 4 real critical points, and only 2 when y is in the unbounded one.
Not all submanifolds X can have as many real critical point as the euclidean distance degree, as can be seen with the parabola, of euclidean distance degree 4 : for all points y, we have at most 3 real critical points. This is illustrated in figure 4.

Figure 4: Examples of real solutions to the EDD equations



We now want to compute the EDD of a smooth complexified algebraic manifold $X \subset \mathbb{C}^m$, defined by polynomials $p_1, ..., p_{m-n}$. We will give two ways of computing it, the first one will involve the degrees of the polynomials $p_1, ..., p_{m-n}$ via a Bezout type argument. The second will use Chern, Segre and polar classes following [14] [5].

If X and y are in generic positions, and because computing the EDD amounts to solving an intersection problem, $\mathrm{EDD}(X)$ can be expressed as the product of the degree of X and the degree of the equation $x - y \in N_x X$. Writing it down in terms of the gradients, one has the following result

Theorem 8. *[4] If $X \subset \mathbb{P}^m$ is a generic complete intersection, defined by polynomials $p_1, ..., p_{m-n}$ of degrees $d_1, ..., d_{m-n}$, then*

$$\mathrm{EDD}(X) = d_1 ... d_{m-n} \sum_{i_1 + ... + i_{m-n} \leq n} (d_1 - 1)^{i_1} ... (d_{m-n} - 1)^{i_{m-n}}$$

We now compute the EDD in terms of classes. Our tools for this requiring projective manifolds, we will take the "projective closure" of X and its bundles of interest, as constructed in section 4.1.2. In what follows, we assume that $\bar{X}$ is orthogonally general relative to the canonical embedding $\mathbb{C}^m \hookrightarrow \mathbb{P}^m$ and the canonical isotopic quadric used to define NX. We have the following maps from the euclidean normal bundle $\hat{N}\bar{X}$.

$$\begin{tikzcd}
\mathbb{P}(\hat{N}\bar{X}) \arrow[rr, "\iota = \mathrm{id}_{\bar{X}} \times \pi_2"] \arrow[dr, "\pi_1"'] & & \mathbb{P}(\bar{X} \times \mathbb{C}^{m+1}) \arrow[dl, "\pi_1'"] \\
& \bar{X} &
\end{tikzcd}$$

We now compute the EDD of X. It was defined as the number of elements of $\pi_1(\pi_2^{-1}(y)) \subset \mathbb{C}^m \hookrightarrow \mathbb{P}^m$ for generic y. To find this number we use the $\deg : A_0(\bar{X}) \to \mathbb{Z}$ map, the degree of a finite set being its size. We have

$$\begin{aligned}\mathrm{EDD}(X) &= \deg([\pi_1(\pi_2^{-1}(y))]) \\ &= \deg([\pi_1(\iota^{-1}(\bar{X} \times \{y\}))])\end{aligned}$$

Denote $\xi = c_1(\mathcal{O}_{\mathbb{P}(\bar{X}\times\mathbb{C}^{m+1})}(1))$. It can be shown that $\xi = [\bar{X} \times H]$, where H is any hyperplane of $\mathbb{P}^m$, using property 1. of Chern classes. Because H can be any hyperplane, one has $\xi^m = [\bar{X} \times \{y\}]$.
We also define $\eta = c_1(\mathcal{O}_{\mathbb{P}(\hat{N}\bar{X})}(1))$, and property 4. of Chern classes gives us $\iota^*(\xi) = \eta$. Rewriting what we had, because π_1 is of degree 1, we obtain

$$\begin{aligned}\mathrm{EDD}(X) &= \deg((\pi_1)_*\iota^*(\xi^m))) \\ &= \deg((\pi_1)_*\eta^m) \\ &= \deg(s_n(\hat{N}\bar{X}))\end{aligned}$$

by definition of Segre classes. To finish, we need some more information about the euclidean normal bundle.

The tautological bundle over $\bar{X}$ is the bundle $\mathcal{O}_{\bar{X}}(-1)$ whose fiber at $[x] \in \bar{X}$ is the line $\mathbb{C}x \subset \mathbb{C}^{m+1}$. By section 4.1.2 we have $\hat{N}\bar{X} \cong N^\infty\bar{X} \oplus \mathcal{O}_{\bar{X}}(-1)$.

Property 3. of Chern classes and the fact that Segre classes are their inverses yields $s(\hat{N}\bar{X}) = s(N^\infty\bar{X})s(\mathcal{O}_{\bar{X}}(-1))$. One can show using the content of section 4.1.2 that $N^\infty\bar{X}$ verifies the following exact sequence

$$0 \to N^\infty\bar{X} \to \bar{X} \times \mathbb{C}^{m+1} \to \hat{T}^*\bar{X} \to 0$$

and from property 3. of Chern classes and the fact that Segre and Chern classes are inverses, we get

$$s(N^\infty\bar{X})s(\hat{T}^*\bar{X}) = s(\bar{X} \times \mathbb{C}^{m+1})$$

For all $k \neq n$, $s_k(\bar{X} \times \mathbb{C}^{m+1}) = 0$, because $\xi^{m+1} = 0$. So $s(\bar{X} \times \mathbb{C}^{m+1}) = 1$, and we get $s(N^\infty\bar{X}) = c(\hat{T}^*\bar{X})$. Lastly, by property 1., one can compute

$$c(\mathcal{O}_{\bar{X}}(-1)) = 1 - c_1(\mathcal{O}_{\bar{X}}(1)) = 1 - \xi$$

and deduce

$$s(\mathcal{O}_{\bar{X}}(-1)) = \sum_{k=0}^{m} \xi^k$$

We then get

$$\begin{aligned}s_n(\hat{N}\bar{X}) &= (c(\hat{T}^*\bar{X})s(\mathcal{O}_{\bar{X}}(-1)))_n \\ &= \sum_{k=0}^{n} c_k(\hat{T}^*\bar{X})\xi^{n-k} \\ &= \sum_{k=0}^{n} p_k(X)\xi^{n-k}\end{aligned}$$

so we conclude

$$\mathrm{EDD}(X) = \sum_{k=0}^{n} \deg_{\mathbb{P}^m}(p_k(X))$$

where $\deg_{\mathbb{P}^m}$ can be defined by the fact that $A(\mathbb{P}^m) = \mathbb{Z}[\xi]/(\xi^{m+1})$.

4.3.2 Bottleneck degree

In this section, our goal will be to study another particular case of the (ACP) equations, where $k = 2$, $X = \mathbb{C}^m$ and $Y \subset \mathbb{C}^m$ is an arbitrary algebraic submanifold. Our goal will be to upper bound the number of such solutions with an expression involving the degree of some classe in a particular Chow ring, which will be called the *bottleneck degree* of Y.

Blow-ups Consider $X \subset \mathbb{C}^m$ a complex variety and $Y \subset X$ a subvariety of X. The blow-up of X at Y is a construction involving of a variety $\mathrm{Bl}_Y(X)$ together with a birationnal map $\pi : \mathrm{Bl}_Y(X) \to X$ such that bl is an isomorphism away from Y but can have non trivial fibers over Y. The algebraic construction of the-blow up is as follows : consider $q_0, ..., q_n$ homogeneous polynomials of the same degree and generating an ideal of saturation $I(Y)$ (the ideal of Y). Then $\mathrm{Bl}_Y(X)$ is the graph of the rationnal maps $[q_0 : ... : q_n]$.

For what follows, we will consider X and Y to be smooth, although it is not strictly necessary. But by restricting to manifolds, we can give a more geometric description (and equivalent definition) of the blow-up. $\mathrm{Bl}_Y(X)$ is made of two kinds of points :

- The usual points $x \in X\backslash Y$
- And the blown-up/special points, of the form (y, v), where $y \in Y$ and $v \in \mathbb{P}(N_yY)$.

The π map sends x to x and (y, v) to y. More precisely, we begin by defining the blow-up of a complex polydisc $B \subset \mathbb{C}^m$ along a coordinate subspace $V = \{z_1 = ... = z_k = 0\}$ as the following manifold

$$\mathrm{Bl}_V(B) = \{((z_1, ..., z_m), [w_1 : ... : w_k : 0 : ... : 0]) \mid \forall i, j, z_i w_j = z_j w_i\} \subset B \times \mathbb{P}^{m-1}$$

We can see that for z outside of V, the second coordinate can only be $[z_1 : ... : z_k : 0 : ... : 0]$, and for $z \in V$, any normal direction to V can be taken, which corresponds to the description given earlier. One important thing to note is that if we have an isomorphism (biholomoric map) $f : B \to B$ taking V to itself, it lifts uniquely into $f : \mathrm{Bl}_V(B) \to \mathrm{Bl}_V(B)$

We generalize this to X and Y by considering a covering of X by open sets U_i, all isomorphic to B via a chart sending Y to a coordinate subspace. We then define the blow-up of X at Y by gluing $X\backslash Y, \mathrm{Bl}_Y(U_i)$ for all i : $X\backslash Y$ and $\mathrm{Bl}_Y(U_i)$ are glued together via $U_i\backslash Y$, and $\mathrm{Bl}_Y(U_i)$ is glued to $\mathrm{Bl}_Y(U_j)$ along the inverse image of $U_i \cap U_j$ via the induced map on blow-ups.

The prototypical example of a blow-up was given when defining the blow up of a polydisc. An important example that will be useful to us, is the blow-up of $X \times X$ along its diagonal Δ_X. $\mathrm{Bl}_{\Delta_X}(X \times X)$ has again two kinds of points, the usual points $(x, x') \notin \Delta_X$ and the special points (x, v) where $x \in X$ and $v \in \mathbb{P}(T_xX)$. Intuitively, the special points give a context where it makes sense to talk about "infinitesimally close points" as they appear when taking limits of sequences of increasingly close points.

One application of blow-ups comes when counting non injective pairs of a function. Suppose $f : M \to N$ is a morphism between algebraic projective manifolds, and that we wish to count the number of tuples $(y_1, y_2) \in M$, $y_1 \neq y_2$ such that $f(y_1) = f(y_2)$. We can frame this as counting the number of elements in $(f \times f)^{-1}(\Delta_N)\backslash\Delta_M$.

To achieve that we define the subvariety $\tilde{D}(f) \subset \mathrm{Bl}_{\Delta_M}(M \times M)$ as follows : if

$$h = (f \times f) \circ \pi : \mathrm{Bl}_{\Delta_M}(M \times M) \to N \times N$$

where $\pi : \mathrm{Bl}_{\Delta_M}(M \times M) \to M \times M$, then $\tilde{D}(f)$ is the residual of $\pi^{-1}(\Delta_X)$ in $h^{-1}(\Delta_Y)$ (cite Fulton). $\tilde{D}(f)$ has two kinds of points

- $(y_1, y_2) \in M\backslash\Delta_M$ such that $f(y_1) = f(y_2)$
- (y, v) where $y \in M$ and $v \in \mathbb{P}(T_yM)$ such that $d_yf(v) = 0$.

Lastly if we denote $\eta = \pi_1 \circ \pi$ where $\pi_1 : M \times M \to M$, we obtain a class in $A(M)$, $\mathbb{D}(f) := \eta_*[\tilde{D}(f)]$ such that its degree gives the number of tuples $(y_1, y_2) \notin \Delta_M$ such that $f(y_1) = f(y_2)$ plus the number of tuples (y, v) where $y \in M$ and $v \in T_yM$ is a vanishing direction of d_yf.

The bottleneck equation and degree Set $X = \mathbb{C}^m$ and Y be an arbitrary algebraic manifold. A *bottleneck* of Y is a solution x of (TCP) when $k = 2$. To study these equations more easily, one studies the *algebraic bottlenecks* of Y, i.e the solutions of (ACP) when $k = 2$. In this situation, (ACP) rewrites as follows

1. $y_1 \neq y_2$
2. $y_1 - y_2 \in N_{y_2}Y$ and $y_2 - y_1 \in N_{y_1}Y$

We would like to find a class in the Chow ring of some variety such that its degree relates to the number of algebraic bottlenecks of Y.

These equations have the issue of involving an open condition, but our tools for solving intersection problems via the Chow ring require the conditions to be algebraic and closed. To solve this, we will relax these equations in order to allow y_1 to be equal to y_2 in a certain sense.

For what follows, we again assume Y to be orthogonally general with respect to the usual hyperplane and quadric at infinity. We will especially make use of the bundle $N^\infty Y$. Recall that for $y \in Y$, $\mathbb{P}(\hat{N}_y\bar{Y}) = \overline{\hat{N}_yY}$. In particular, by the construction

$$\mathbb{P}(N^\infty\bar{Y}) = \overline{\hat{N}_yY} \cap H_\infty$$

We then get that for all $(y, v) \in NY$, $y \in \mathbb{C}^{m+1}$ and $v \neq 0$ are not proportional [7]. With that, we introduce the following map

$$f: \begin{array}{cl} \mathbb{P}(N^\infty\bar{Y}) & \longrightarrow G(2, m+1) \\ (y, v) & \longmapsto \operatorname{span}\{y, v\} \end{array}$$

where $G(2, m+1)$ is the Grassmannian of lines in $\mathbb{P}^m$ and the span of y and v is the unique projective line passing through both points. Note that we can rephrase the algebraic bottleneck equations with f : for $y_1 \neq y_2 \in \bar{Y}$, there exists $v_1 \in \mathbb{P}(N^\infty_{y_1}\bar{Y})$ and $v_2 \in \mathbb{P}(N^\infty_{y_2}\bar{Y})$ such that $f(y_1, v_1) = f(y_2, v_2)$ iff (y_1, y_2) is a solution to the algebraic bottleneck equations. Moreover, v_1, v_2 are uniquely determined by y_1, y_2.Hence counting the number of algebraic bottlenecks amounts to counting non injective pairs of f.

By the previous discussion, we can relate the number of algebraic bottlenecks of $\bar{Y}$ to $\deg \mathbb{D}(f)$. This number then counts

- Tuples $(y_1, v_1), (y_2, v_2)$ such that $f(y_1, v_1) = f(y_2, v_2)$ i.e algebraic bottlenecks of $\bar{Y}$
- Points (y, v, w), where $(y, v) \in \mathbb{P}(N^\infty\bar{Y})$ and w a tangent line to $N_\infty\bar{Y}$ at (y, v) such that $d_{(y,v)}f(w) = 0$.

The following proposition allows us to give a geometric interpretation of the second kind of points. Note that, denoting $p : \mathbb{C}^{m+1}\backslash\{0\} \to \mathbb{P}^m$ the usual projection, we have

$$T_{(y,v)}\mathbb{P}(N^\infty\bar{Y}) \cong T_y\bar{Y} \oplus d_vp(N^\infty_y\bar{Y})$$

Proposition 12. *Let $(y, v) \in \mathbb{P}(N^\infty\bar{Y})$ such that $y \in Y \subset \mathbb{C}^m$. We have*

$$\ker d_{(y,v)}f = \{\lambda v \mid \lambda v \in Q\} \oplus 0 \subset T_{(y,v)}\mathbb{P}(N^\infty\bar{Y})$$

Proof. We start by recalling how the canonical embedding $\mathbb{C}^m \hookrightarrow \mathbb{P}^m$ is defined : we start by embedding $\mathbb{C}^m \hookrightarrow \mathbb{C}^m \times \{1\}$ then taking the projection. Hence $y \in Y \subset \mathbb{C}^{m+1}$ can be written as $y = (\tilde{y}, 1)$. With that in mind and denoting $(NY)_0$ the image of the zero section, we can define the following projection

$$g: \begin{array}{cl} NY\backslash(NY)_0 & \longrightarrow \mathbb{P}(N^\infty\bar{Y}) \\ (y, v) & \longmapsto ([y], [\tilde{v} : 0]) \end{array}$$

as v can be written as $v = (\tilde{v}, *)$ where $\tilde{v}$ is a normal vector to Y seen in $\mathbb{C}^m$. If we write $V(2, m+1) \subset \mathcal{M}_{m+1,2}(\mathbb{C})$ the submanifold of matrices of rank 2, we get the following commutative diagram

$$\begin{array}{ccc} NY\backslash(NY)_0 & \xrightarrow{\tilde{f}} & V(2, m+1) \\ \Big\downarrow g & & \Big\downarrow q \\ \mathbb{P}(N^\infty\bar{Y}) & \xrightarrow{f} & G(2, m+1) \end{array}$$

where q is the canonical projection and $\tilde{f} : (y,v) \mapsto (y \mid v)$. This is well defined as y and v are not proportional.

Let $w \in \ker d_{(y,[v])} f$ where $(y,[v]) \in \mathbb{P}(N^\infty \bar{Y})$. Because g is a surjective submersion, we find $y \in Y$ and $v \in N_y Y$ of the form $v = (\tilde{v}, 0)$, and

$$u := u_t + u_n \in T_{(y,v)}(NY) \cong T_y Y \oplus N_y Y$$

such that $w = d_{(y,v)} g(u_t + u_n)$. It now suffices to show that u_n and u_t are proportional to v and u_t is in the isotopic quadric.

$\tilde{f}$ being an embedding, we see u as an element of $V(2, m+1)$. w being in the kernel of df at $(y,[v])$, u is in the kernel of dq at $(y \mid v)$, so it is of the form $(y \mid v)M$ where $M \in \mathcal{M}_2(\mathbb{C})$. u_t and u_n being respectively tangent and normal to $Y \subset \mathbb{C}^{m+1}$, we have

$$u = \begin{pmatrix} \tilde{u}_t & \tilde{u}_n \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \tilde{y} & \tilde{v} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

We easily get $a = 0$, $b = \lambda$. Then $\lambda \tilde{y} + d\tilde{v} = \tilde{u}_n$ and so $\lambda \tilde{y} = \tilde{u}_n - d\tilde{v}$ is a normal vector to Y if seen in $\mathbb{C}^m$. By the bottleneck general condition, λ must be 0, so u_n is proportional to v. Lastly, $c\tilde{v} = \tilde{u}_t$ which is in $N_y Y$ and $T_y Y$ if seen in $\mathbb{C}^m$, so this vector is in the isotopic quadric. We showed what we wanted.

One can check by similar computations that if u_t and u_n verify such conditions, w is in the kernel of df at $(y,[v])$. $\square$

With the help of this result, we see that the second kind of points counted in $\mathbb{D}(f)$ are a purely complex phenomenon, as the isotopic quadric is not reduced to 0 thanks to the complex structure.

Thanks to the previous discussion, we can define the *bottleneck degree* of $\bar{Y}$ as $\mathrm{BND}(\bar{Y}) = \deg \mathbb{D}(f)$. Because the points y_1, y_2 might lie in Y_∞, this doesn't give us the bottleneck degree of Y. However one can fix that in the following way, as $Y_\infty \subset \mathbb{P}^{m-1}$

Definition 11 (Affine bottleneck degree)**.** *If $Y \subset \mathbb{C}^m$ is orthogonally general, we define its bottleneck degree as*

$$\mathrm{BND}(Y) = \mathrm{BND}(\bar{Y}) - \mathrm{BND}(Y_\infty)$$

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